# Dancing in the Dark: Sentiment Shocks and Economic Activity*

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**Abstract**

Expectations—some justified, some not—are widely believed to drive business cycle fluctuations. Yet the conditions under which the economy becomes susceptible to noise-driven shifts in expectations ("sentiment shocks") remain poorly understood. We show that the macroeconomic effects of such shocks are strongly state dependent. Standard signal-extraction logic implies that expectations and their dispersion become more sensitive to noise as uncertainty increases. Consistent with this implication, U.S. time-series evidence shows that when disagreement is low, sentiment shocks have little effect on economic activity and are largely absorbed by prices, whereas when disagreement is high, they have sizable real effects with little impact on prices. We account for this state-dependent transmission in an analytical New Keynesian model with dispersed information, in which agents infer fundamentals from noisy signals—effectively dancing in the dark.

**Keywords:** sentiment shocks, noise shocks, animal spirits, business cycles, nowcast errors, disagreement, dispersed beliefs

**JEL classification:** C32, C34, D84, E21, E23, E32

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# 1. Introduction

Economic fluctuations are, to no small extent, caused by non-fundamental shocks. These shocks come with various labels, such as noise, animal spirits, or sentiment shocks, yet the underlying notion is similar. It traces back to Pigou (1927) and Keynes (1936) and has been substantiated more recently in quantitative work (for instance, Blanchard et al., 2013; Angeletos and La'O, 2013; Lagerborg et al., 2023). Ultimately, such shocks matter because economic actions are based on expectations, which, in turn, are prone to errors or coordination failures and may even become self-fulfilling. In what follows, we refer to this class of shocks as "sentiment shocks". Crucially, their effects are likely to depend on the dispersion of expectations: when views about the economic outlook diverge, economic activity may be more sensitive to sentiment shocks than when expectations are closely aligned.

In this paper, we thus ask how such disagreement shapes the economic impact of sentiment shocks. Our identification strategy follows Enders et al. (2021) and is centered around the nowcast error for output growth. Sentiment shocks induce a negative co-movement between the nowcast error and output: This sets them apart from fundamental shocks and rationalizes the sign restrictions we impose on a VAR model that we estimate on U.S. time-series data. The key contribution of this paper is to allow the effects of sentiment shocks to vary with forecaster disagreement. We find that disagreement is indeed crucial for how sentiment shocks play out: in normal times, a sentiment shock has no effect on economic activity and is fully absorbed by rising prices. In contrast, during periods of high disagreement, economic activity responds strongly while prices remain unchanged.

We rationalize the evidence in a stylized noisy-information model à la Lorenzoni (2009). In the model, aggregate technology is not directly observed—expectations are formed while "dancing in the dark"—and forecasts are based on private information and public signals. If aggregate technology becomes more volatile and hence more uncertain, weight shifts from priors to signals, increasing the responsiveness of expectations to both public noise—which captures sentiment shocks in the model—and idiosyncratic information, thereby generating greater disagreement. The model thus predicts, consistent with the evidence, that greater disagreement is associated with stronger responses of economic activity to sentiment shocks.

More in detail, our empirical analysis is based on a two-step approach. In the first step, we identify sentiment shocks using a bivariate VAR model that incorporates quarterly observations of the nowcast error and output over the 50-year period from

1969 to 2019. The nowcast error is defined as the difference between actual output growth and the median estimate reported in the Survey of Professional Forecasters (SPF) in real time. As a measure of real-time misperceptions, it provides us—ex post—with an informational advantage that is key to identifying sentiment shocks (Blanchard et al., 2013).

Importantly, the nowcast errors are not only due to sentiment shocks; they may just as well reflect fundamental shocks. Still, we use the nowcast error to identify sentiment shocks, which are distinctive in causing a negative co-movement between output and the nowcast error. For example, an expansionary shock due to an unexpected change in total factor productivity generates a positive nowcast error: output expands and turns out higher than expected, resulting in a positive nowcast error.[1] In contrast, a favorable sentiment shock leads expected current growth to overshoot actual growth (resulting in a negative nowcast error) while simultaneously providing a boost to economic activity. Building on this insight, we use sign restrictions to identify sentiment shocks (Benhima and Poilly, 2021; Chahrour et al., 2021; Enders et al., 2021).

We move beyond existing work on the effects of sentiment shocks by employing a smooth-transition VAR model, which allows the effects of shocks to vary with the degree of disagreement among forecasters—measured by the cross-sectional interquartile range of individual output growth nowcasts in the SPF. Based on the empirical cumulative distribution function within our sample, we define two polar regimes of high and low disagreement, allowing for a smooth transition between them. In the second step, we use the sentiment shocks identified in the VAR to estimate their effects using state-dependent local projections. This approach allows us to conveniently expand the set of dependent variables while consistently accounting for regime dependence, both in the identification and in the estimation of their transmission.

The central result of our analysis is that the effect of sentiment shocks differs depending on the level of disagreement among forecasters. When disagreement is high, a sentiment shock leads to a significant increase in output that persists for about four to five years. At the same time, sentiment shocks have virtually no effect on prices. In terms of magnitude, we find a sizable effect: a sentiment shock that implies an overprediction of current output growth by one percentage point increases output by two percent after two years. In contrast, when disagreement is low, an

[1] This feature is not specific to productivity shocks but is a general property of non-sentiment or fundamental shocks. Furthermore, the effects are symmetric: A generic contractionary shock induces a decline in output and, at the same time, a negative nowcast error if the shock is not fully observed in real time. The assumption of symmetry extends to sentiment shocks as well.

expansionary sentiment shock has little impact on output. Instead, it is absorbed by rising prices, which increase by approximately 1.5 percent after two to three years before gradually returning to their pre-shock level.

We find that these outcomes are robust to various modifications of the analysis and detect similar patterns across a range of macroeconomic indicators. Consumption and investment increase in response to sentiment shocks when disagreement is high but remain unresponsive or even decline when disagreement is low. Monetary policy reacts to rising prices following a sentiment shock by raising the federal funds rate in the low-disagreement regime, which helps explain the (mildly) recessionary effect in this case. In an extension, we establish similar patterns in the responses to news shocks about future total factor productivity: they are strongly expansionary only when disagreement is high. Furthermore, we show that our approach can be cross-validated with measures of economic uncertainty, which, according to our stylized model, also rise when aggregate technology becomes more volatile. Furthermore, the state-dependent pattern remains robust when we control for SPF forecast revisions and identified technology news shocks. Finally, the main results are insensitive to expanding the information set used to identify sentiment shocks.

We explain the nature of sentiment shocks and how their impact on macroeconomic outcomes varies with the prevailing level of fundamental uncertainty within a version of the dispersed information model of Lorenzoni (2009). In the model, higher fundamental uncertainty also generates greater disagreement, providing a direct link between the mechanism and our empirical conditioning variable. A key feature of the model is that households and firms do not observe aggregate productivity at the time of decision-making. Instead, they rely on expectations, or more specifically, nowcasts, which they form based on a public signal and private information, extracted in turn from either their own productivity (firms) or observed prices (households).

Assuming that prices are predetermined (rather than staggered as in the original model), the model can be solved in closed form. In the model, firms overestimate aggregate productivity and, therefore, expect competitors' prices to be lower following an expansionary sentiment shock, leading them to reduce their own prices. Under a mild condition, this boosts output while markups decline. At the same time, households expect better fundamentals and increase consumption accordingly. We then show that the response to sentiment shocks varies with the level of disagreement. Intuitively, the optimal weights placed on signals depend on the perceived uncertainty

about aggregate technology.[2] When the volatility of aggregate productivity is (perceived to be) high, expectations become more responsive to signals and, thus, more sensitive to the noise in the public signal, which corresponds to aggregate sentiment shocks.

The paper is organized as follows. The remainder of the introduction places the paper in the context of the literature, clarifying its contribution. The next section uses a stylized setup to fix ideas and set the stage for the empirical analysis, developed in Section 3. Section 4 reports the empirical results. Section 5 rationalizes the findings using the dispersed information model. The final section concludes.

**Related Literature.** Our paper relates to several strands of the literature. First, a large number of studies have examined how uncertainty affects the transmission of shocks, with particular emphasis on the effects of fiscal and monetary policy. A robust finding is that policy measures tend to be less effective when uncertainty is high, typically measured using conventional proxies of macroeconomic or financial uncertainty (Aastveit et al., 2017; Castelnuovo and Pellegrino, 2018; Hauzenberger et al., 2021). Uncertainty, and the state of the economy more broadly, potentially also shape how expectations respond to shocks, and hence the transmission of uncertainty and TFP shocks (Ilut and Schneider, 2014; Lhuissier and Tripier, 2021; Gambetti et al., 2023; Bianchi et al., 2024; Antonova et al., 2026). Uncertainty also matters for price setting (Baley and Blanco, 2019; Bachmann et al., 2019). More generally, disagreement and uncertainty are distinct concepts and need not move together, as emphasized by Zarnowitz and Lambros (1987), Born et al. (2023), and Zohar (2024). Under our structural interpretation, disagreement is an endogenous outcome of the information environment rather than a direct measure of uncertainty.

Like us, Ricco et al. (2016), Falck et al. (2021) and Dong et al. (2025) highlight state-dependent effects based on disagreement. However, their focus is on fiscal and monetary policy, and they find that lower levels of disagreement give rise to stronger effects, possibly due to more effective policy communication. In contrast, we study the conditions under which non-fundamental shocks unfold. Our contribution is to show how disagreement shapes the transmission of sentiment shocks, providing both new evidence and a theoretical account.

This question is particularly relevant in light of a second strand of the literature that seeks to assess the importance of news and "noise" for business cycle fluctu-

[2] And indeed, Benamar et al. (2020) find that investors pay more attention to (macroeconomic) data when uncertainty is high. See also Figure 1 below.

ations, with partly conflicting results (Beaudry and Portier, 2006; Beaudry et al., 2011; Schmitt-Grohé and Uribe, 2012; Barsky et al., 2015; Benhima and Poilly, 2021). These differences likely arise because distinguishing news from noise remains challenging (Barsky and Sims, 2011; Chahrour and Jurado, 2018; Levchenko and Pandalai-Nayar, 2020). Against this background, it is important to condition the effects of sentiment shocks—broadly understood—on the degree of disagreement, as disagreement reflects the consequences of information frictions that give rise to sentiment shocks in the first place.

Finally, our work connects to the literature on how public signals shape coordination and welfare. Seminal work by Morris and Shin (2002) and Angeletos and Pavan (2007) shows how public information acts as a coordination device in environments with strategic complementarities. Follow-up work in New-Keynesian settings yields mixed welfare implications (Hellwig, 2005; Walsh, 2007; Ehrmann and Fratzscher, 2007; Cornand and Heinemann, 2008), and Angeletos et al. (2016) stress that the answer depends on the source of aggregate fluctuations. We contribute to this strand of the literature by providing evidence that coordination problems actually shape economic outcomes.

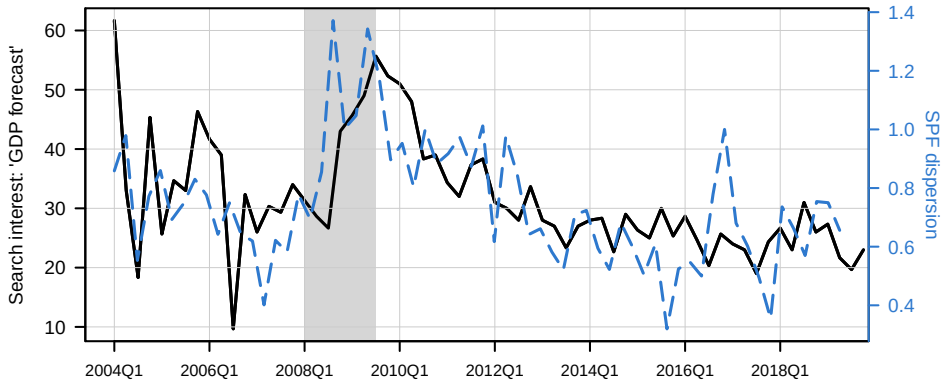
## 2. Fixing ideas

To set the stage for the empirical analysis, we fix ideas and sketch why, in theory, fundamental uncertainty matters for how sentiment shocks unfold. Specifically, we zoom in on the signal-extraction problem, which is central to the general-equilibrium model in Section 5 below. Firm productivity features both an aggregate and an idiosyncratic component. In real time, firms do not directly observe aggregate technology, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, where $\sigma_\varepsilon^2$ measures fundamental uncertainty. Instead, each firm receives a private signal $a_i = \varepsilon + \eta_i$, which contains idiosyncratic noise $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$. By construction, idiosyncratic noise averages out in the aggregate. Firms also observe a public signal, $s = \varepsilon + e$, which is noisy as well, with $e \sim \mathcal{N}(0, \sigma_e^2)$. The disturbance $e$ represents the noise shock, the effects of which we seek to identify below. In this context, the optimal estimate of aggregate technology is given by

$$\mathbb{E}_i[\varepsilon|s, a_i] = \rho s + \delta a_i = (\rho + \delta)\varepsilon + \rho e + \delta \eta_i,$$

$$\text{with} \qquad \rho = \frac{\sigma_\eta^2}{\sigma_e^2 + \sigma_\eta^2 + \sigma_e^2\sigma_\eta^2/\sigma_\varepsilon^2} \qquad \text{and} \qquad \delta = \frac{\sigma_e^2}{\sigma_e^2 + \sigma_\eta^2 + \sigma_e^2\sigma_\eta^2/\sigma_\varepsilon^2}.$$

**Figure 1:** Attention to public signals.




*Notes*: The figure shows Google searches for the term 'GDP forecast' (black solid line, left axis) and the interquartile range across forecasters in the SPF regarding output growth forecasts four quarters ahead (blue dashed line, right axis). Gray bars denote the NBER recession dates.

We focus on variation in fundamental uncertainty $\sigma_\varepsilon^2$, motivated by evidence of stochastic volatility in U.S. aggregate TFP innovations.[3] The impact of both signals on expectations increases in $\sigma_\varepsilon^2$: if uncertainty is high, firms pay more attention to the signals instead of relying on their prior (zero in this case). This has two implications. First, the impact of noise on expectations rises—via a higher $\rho$. Second, the dispersion of expectations, given by $D = \delta^2\sigma_\eta^2$, and economic uncertainty, given by the agent's forecast error variance $FEV = \rho\sigma_e^2$, increase. Fundamental uncertainty thus provides a mechanism through which disagreement and economic uncertainty move together.[4] This motivates our choice for the baseline state variable—expectation dispersion—and the cross-validation using other measures of economic uncertainty.

The data in Figure 1 support the notion that expectations are more susceptible to signals during times of greater forecaster disagreement. Over time, disagreement among SPF participants (measured by the interquartile range) co-moves strongly with Google search interest in the term "GDP forecast," which may proxy for attention to public signals. This pattern lends support to conditioning the effects of sentiment shocks on disagreement in the empirical analysis below.

[3] In a stochastic volatility model estimated on the TFP series by Fernald (2014), we find that the conditional volatility of aggregate TFP innovations shows substantial time variation, consistent with Born and Pfeifer (2014).

[4] Disagreement may also arise from changes in the precision and composition of signals. In Appendix C, we show that increasing noise in private signals raises sensitivity to sentiment shocks, but raises disagreement only if private signals are initially sufficiently precise, a case we consider unlikely (Comin and Philippon, 2005; Lorenzoni, 2009; Andrade et al., 2022). Increasing noise in the public signal or shifting noise toward it can raise disagreement but reduces sensitivity to sentiment shocks.

How expectations respond to signals matters for economic activity and prices because expectations feed back into the decisions of households and firms: In Section 5, we derive how expectations map into macroeconomic outcomes. We show formally that greater fundamental uncertainty increases the dispersion of expectations and amplifies (or attenuates) the effects of sentiment shocks on output (prices).[5] The model also sheds light on the mechanisms underlying this mapping. While the signal-extraction problem is central, the key features at the heart of the New Keynesian framework turn out to be essential as well.

## 3. Empirical framework

This section presents our empirical framework, explaining how we identify sentiment shocks in a non-linear setting and how we estimate state-dependent local projections. We allow the sentiment shocks to depend on the level of disagreement. For that, we define two polar regimes of disagreement and discuss the regime allocation based on the disagreement series. Before doing that, we introduce the data.

### 3.1 Data

We use quarterly data for the US, ranging from 1969Q4 to 2019Q4, for a set of macroeconomic quantities: output, nowcast errors of output, the dispersion of nowcast errors, consumer prices, the federal funds rate, the S&P 500 index, and various sub-components of output. We use final-release data from the Bureau of Economic Analysis (BEA) for real gross domestic product to measure output. Nowcast errors of GDP growth are computed as the difference between BEA's actual first-release output growth rate and the equivalent SPF survey nowcast.[6] We measure forecaster disagreement as the dispersion of nowcasts by the interquartile range of the partici-

[5] Note that it does not matter whether this is a real or merely perceived change. In a similar vein, Gemmi and Mihet (2023) investigate the impact of uncertainty on household expectations about inflation, distinguishing between uncertainty due to higher volatility of the fundamental or due to higher volatility of the signals (noise).

[6] For the SPF nowcasts of output, we use the series DRGDP2, which we obtain from the Real-time Data Research Center of the Philadelphia Fed. This series corresponds to the median nowcast of the quarterly growth rate of real output, seasonally adjusted at annual rates (real GNP before 1992 and real GDP afterwards). Prior to 1981Q3, the SPF asked for nominal GNP only. In this case, the implied nowcast for real GNP is computed based on the nowcast for the price index of GNP. We investigate only the median nowcast, not nowcasts in the tails of the distribution, as those produce similar responses (Boeck and Pfarrhofer, 2025). Using final-release data to construct nowcasts generally yields very similar results (see Enders et al., 2021).

pants' forecasts for current-quarter output growth (in percentage points) in the SPF.[7] As our stylized model in Section 2 predicts that economic uncertainty rises alongside disagreement, we further examine the robustness of our results using various measures of economic uncertainty. Disagreement and uncertainty are conceptually distinct objects (Zohar, 2024), although disagreement has frequently been used as a proxy for economic uncertainty (e.g. Bachmann et al., 2013; Ilut and Schneider, 2014; Bianchi et al., 2024). The remaining variables are relatively standard, and we provide further details in Appendix A.

### 3.2 Non-linear identification of sentiment shocks

In terms of identification, we build on Enders et al. (2021) and the Bayesian variant in Chahrour et al. (2021). However, we move beyond the linear VAR framework to allow for state-dependent effects. Specifically, we estimate a bivariate Bayesian smooth-transition vector autoregressive (STVAR) model, which we identify using sign restrictions. The $2 \times 1$ vector of endogenous variables $\boldsymbol{y}_t = (ne_t, gdp_t)'$ comprises the nowcast error of output growth and the actual growth rate of GDP. We capture state dependence using a transition function $F(z_{t-1}) \in [0, 1]$, which reflects the probability of being in the high-disagreement regime at time $t-1$. Let the time series process $\{\boldsymbol{y}_t\}_{t=1}^{T}$ follow

$$\begin{aligned} \boldsymbol{y}_t = &(\boldsymbol{c}_{11} + \boldsymbol{A}_{11}\boldsymbol{y}_{t-1} + \ldots + \boldsymbol{A}_{1p}\boldsymbol{y}_{t-p}) \times F(z_{t-1}) \\ &+ (\boldsymbol{c}_{21} + \boldsymbol{A}_{21}\boldsymbol{y}_{t-1} + \ldots + \boldsymbol{A}_{2p}\boldsymbol{y}_{t-p}) \times (1 - F(z_{t-1})) \\ &+ \boldsymbol{c}_2 t + \boldsymbol{c}_3 t^2 + \boldsymbol{S}_t \boldsymbol{\varepsilon}_t, \qquad\qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}), \end{aligned} \tag{3.1}$$

where $\boldsymbol{A}_{rj}$ are $n \times n$ coefficient matrices for regime $r \in \{1, 2\}$ and lag $j \in \{1, 2, \ldots, p\}$, the $2 \times 1$ vectors $\boldsymbol{c}_{r1}$, $\boldsymbol{c}_2$, and $\boldsymbol{c}_3$ denote the coefficients corresponding to the intercept, trend, and quadratic trend. The $2 \times 1$ vector $\boldsymbol{\varepsilon}_t$ denotes the structural errors, which are normally distributed with zero mean and unit variance.[8] Hence, $\boldsymbol{S}_t$ denotes the structural impact matrix, which is time-varying due to its state-dependence. This implies that the reduced-form errors $\boldsymbol{u}_t = \boldsymbol{S}_t \boldsymbol{\varepsilon}_t$ follow a Gaussian distribution with

[7] Accordingly, we construct our measure from the same dataset that provides the expectation data. As alternative specifications, we also consider the standard deviation across forecasts and other measures of economic uncertainty, which are discussed in greater detail in the robustness section below.

[8] Without loss of generality, we define the second row of $\boldsymbol{\varepsilon}_t$ to be a sentiment shock, which we use as an exogenous shock in the local projection (3.5) below.

zero mean and the $2 \times 2$ covariance matrix $\boldsymbol{\Sigma}_t = \boldsymbol{S}_t \boldsymbol{S}_t'$. To make the state-dependence explicit, we write

$$\boldsymbol{\Sigma}_t = \tilde{\boldsymbol{\Sigma}}_1 F(z_{t-1}) + \tilde{\boldsymbol{\Sigma}}_2 (1 - F(z_{t-1})). \tag{3.2}$$

This leads to $\tilde{\boldsymbol{\Sigma}}_r = \tilde{\boldsymbol{S}}_r \tilde{\boldsymbol{S}}_r'$, where $\tilde{\boldsymbol{S}}_r$ $(r = 1, 2)$ is the regime-specific structural impact matrix. We discuss further details on the Bayesian estimation of the STVAR model in Appendix A.2.

To estimate the state-dependent effects within the STVAR model, we interact the coefficients with the transition function $F(z_{t-1}) \in [0, 1]$, which reflects the weight of being in the high-disagreement regime at time $t - 1$. This specification reflects the fact that disagreement is not binary; rather, it can vary in intensity over time. In our estimation, we leverage this continuous variation to assess how the effect of sentiment shocks depends on forecaster disagreement. The concept of a regime is used solely to define limiting cases for illustrative purposes.

The specification of the transition function involves two steps: the choice of the indicator and the specification of the mapping of the indicator into weights. First, we identify the regimes with the level of disagreement about GDP growth nowcasts, as captured by the variable $z_{t-1}$.[9] Second, we specify the transition function on the basis of the empirical cumulative distribution function (CDF) in our sample, adopting the approach of Born et al. (2020) to our setting.[10] Formally, we have

$$F(z_{t-1}) = \frac{1}{T} \sum_{j=1}^{T} \mathbb{1}(z_j \leq z_{t-1}), \tag{3.3}$$

where $T$ is the number of observations in our sample, $\mathbb{1}$ is an indicator function, and $j$ indexes all observations. The function equals one if disagreement is at the sample maximum: a situation in which information is extremely dispersed relative to the rest of the sample. Instead, the function equals $1/T$ at the sample minimum. Values approaching zero represent the limiting low-disagreement case. As disagreement is continuous, the economy is hardly ever in one of these two polar regimes. This is captured in the estimation, as each observation is a weighted average of the dynamics in the two regimes.

The identification of the sentiment shock builds upon nowcast errors, which are an ex-post measure of the mistakes made by market participants. Incorporating these nowcast errors into the model provides us with an informational advantage over mar-

[9] We postpone the exact details on the construction and discussion of this indicator to Section 3.4.

[10] In contrast to the literature building on Auerbach and Gorodnichenko (2012), this allows us to avoid imposing a specific parametric transition function. Studies in this tradition (see, for instance, Auerbach and Gorodnichenko, 2013, Caggiano et al., 2014, Tenreyro and Thwaites, 2016, or Falck et al., 2021) use the logistic function as the transition function and calibrate rather than estimate the involved smoothness parameter. We follow their calibration strategy to provide robustness for our baseline results using the logistic function and report the results in Figure B2 in the appendix.

ket participants. Identifying sentiment shocks without such an advantage is infeasible (Blanchard et al., 2013). Specifically, identification is achieved through sign restrictions (Rubio-Ramirez et al., 2010). For the non-sentiment shock, we restrict the signs such that there is a positive comovement between the nowcast error and GDP growth. In the case of a fundamental shock, the shock is generally not fully processed in real time (Coibion and Gorodnichenko, 2012; Coibion and Gorodnichenko, 2015), such that its effects exceed expectations. In contrast, a sentiment shock is characterized by excess movements of expectations. Since expectations overshoot actual output growth (positively or negatively), the nowcast error has the opposite sign to that of actual output growth. Hence, for the sentiment shock, we impose a negative comovement.[11] Specifically, we assume that

$$\begin{bmatrix} u_t^{ne} \\ u_t^{gdp} \end{bmatrix} = \begin{pmatrix} + & + \\ + & - \end{pmatrix} \begin{bmatrix} \varepsilon_t^{ns} \\ \varepsilon_t^{s} \end{bmatrix}, \tag{3.4}$$

where $\varepsilon_t^{ns}$ and $\varepsilon_t^{s}$ denote the *non-sentiment* and *sentiment* shocks, respectively. Importantly, while the sign restrictions apply in both disagreement regimes, we allow the structural impact matrix to vary across regimes. Thus, we account for state dependence in the identification of the sentiment shock itself, rather than introducing it only when estimating its effects. We recover the sentiment shock as $\varepsilon_t^s = \boldsymbol{e}_2' \boldsymbol{S}_t^{-1} \boldsymbol{u}_t$, where $\boldsymbol{e}_2$ denotes a unit vector with its second entry equal to one. Sign restrictions only provide set identification, and as such, the framework inherently captures both the uncertainty regarding the imposed sign restrictions and the uncertainty regarding the estimated parameters. We generate the shocks by using the arithmetic average across both types of uncertainty. We describe the procedure in more detail in Appendix A.2.

We use a two-step approach: we first identify the sentiment shock in a parsimonious STVAR and then estimate its effects on a broader set of macroeconomic indicators using state-dependent local projections. This approach allows us to examine additional outcomes while keeping the identified shock series fixed. By contrast, adding these variables to the STVAR would change the identified shock series and would necessitate additional identification assumptions. Recovering the structural shock, nevertheless, requires a sufficiently informative set of variables. Following Forni and Gambetti (2014), Appendix B.3 assesses whether lagged information from a high-dimensional macroeconomic dataset predicts the identified sentiment shocks.

[11] Using a richer specification, Benhima and Poilly (2021) further distinguish between *supply noise* and *demand noise* shocks based on expectation errors in output growth and inflation.

We show that our main results remain robust to expanding the STVAR's information set.

### 3.3 State-dependent local projections

To study the transmission of sentiment shocks to a broader set of variables, we use state-dependent local projections following Jordà (2005). We use the same lagged disagreement indicator and the same smooth-transition function as in the STVAR. Both stages, therefore, use identical regime weights, accounting for state dependence when identifying the shocks and subsequently estimating their effects on a broader set of macroeconomic variables.

This results in smooth-transition local projections (STLP). Letting $y_{t+h}$ denote the response of a particular variable at time $t+h$ to a sentiment shock $\varepsilon_t^s$ at time $t$, we consider a model that depends on the level of disagreement. It reads:

$$\begin{aligned}\Delta^h y_{t+h} = &\left(\alpha_h^L + \beta_h^L \varepsilon_t^s + \boldsymbol{X}_{t-1}'\boldsymbol{\gamma}_h^L\right) \times \left(1 - F(z_{t-1})\right) \\ &+ \left(\alpha_h^H + \beta_h^H \varepsilon_t^s + \boldsymbol{X}_{t-1}'\boldsymbol{\gamma}_h^H\right) \times F(z_{t-1}) \\ &+ \tau_{1h} t + \tau_{2h} t^2 + u_{t+h}^{(h)}, \qquad\qquad u_{t+h}^{(h)} \sim \mathcal{N}(0, \sigma_h^2),\end{aligned} \tag{3.5}$$

where $\Delta^h y_{t+h} = y_{t+h} - y_{t-1}$ denotes cumulative differences and $h = 0, \ldots, H$ denotes the number of periods after the shock hits the economy. The coefficient of interest is $\beta_h^r$, which denotes the causal, state-dependent response at horizon $h$ to a sentiment shock, where $r \in \{L, H\}$ refers to the low-disagreement ($L$) and high-disagreement ($H$) regimes, respectively.[12]

The model specification includes a linear-quadratic trend ($\tau_{1h}$ and $\tau_{2h}$), state-dependent constants ($\alpha_h^r$), and state-dependent coefficients ($\boldsymbol{\gamma}_h^r$) for the vector of control variables $\boldsymbol{X}_{t-1}$. The $n_x \times 1$ vector $\boldsymbol{X}_{t-1}$ contains lagged control variables. For the transition function $F(z_{t-1})$, we use the same specification as before. The regression residual is denoted by $u_{t+h}$ and is distributed as Gaussian with zero mean and $\sigma_h^2$ variance. To address the issue of potential autocorrelation in the residuals, we apply the strategy proposed by Lusompa (2023).[13] In all specifications, we use four lags of the shock series and the endogenous variable in the regression as controls, which robustifies inference (Montiel-Olea and Plagborg-Møller, 2021). The specification in

[12] This model nests a linear specification in which we suppress the state-dependency. The equation then reads as follows: $\Delta^h y_{t+h} = \alpha_h + \beta_h \varepsilon_t^s + \boldsymbol{X}_{t-1}'\boldsymbol{\gamma}_h + \tau_{1h} t + \tau_{2h} t^2 + u_{t+h}^h$ with $u_{t+h}^h \sim \mathcal{N}(0, \sigma_h^2)$.

[13] Lusompa (2023) shows that the autocorrelation process of LPs can be written as a vector moving average process and can be corrected using a consistent GLS estimator. This strategy exploits the fact that, at least, the horizon-1 LP residuals are not autocorrelated.

long differences shows considerable small sample gains when the impulse response of interest is estimated using an externally identified shock (Piger and Stockwell, 2023). We estimate the local projections in a Bayesian framework, which allows us to impose regularization techniques on the vector of coefficients corresponding to the control variables (Carvalho et al., 2010). We provide further details on the Bayesian estimation of the STLP in Appendix A.3.

For the estimation of impulse responses, the parameter $\beta_h^r$ ($r \in \{L, H\}$) provides a direct causal estimate of the response of the dependent variable to the sentiment shock $\varepsilon_t^s$. We investigate the two polar cases in which the economy is in a low- or high-disagreement regime today, indexed by $z_{t-1}$. Formally, we have

$$\left.\frac{\partial \Delta^h y_{t+h}}{\partial \varepsilon_t^s}\right|_{z_{t-1}} = \beta_h^L \times \Big(1 - F(z_{t-1})\Big) + \beta_h^H \times F(z_{t-1}).$$

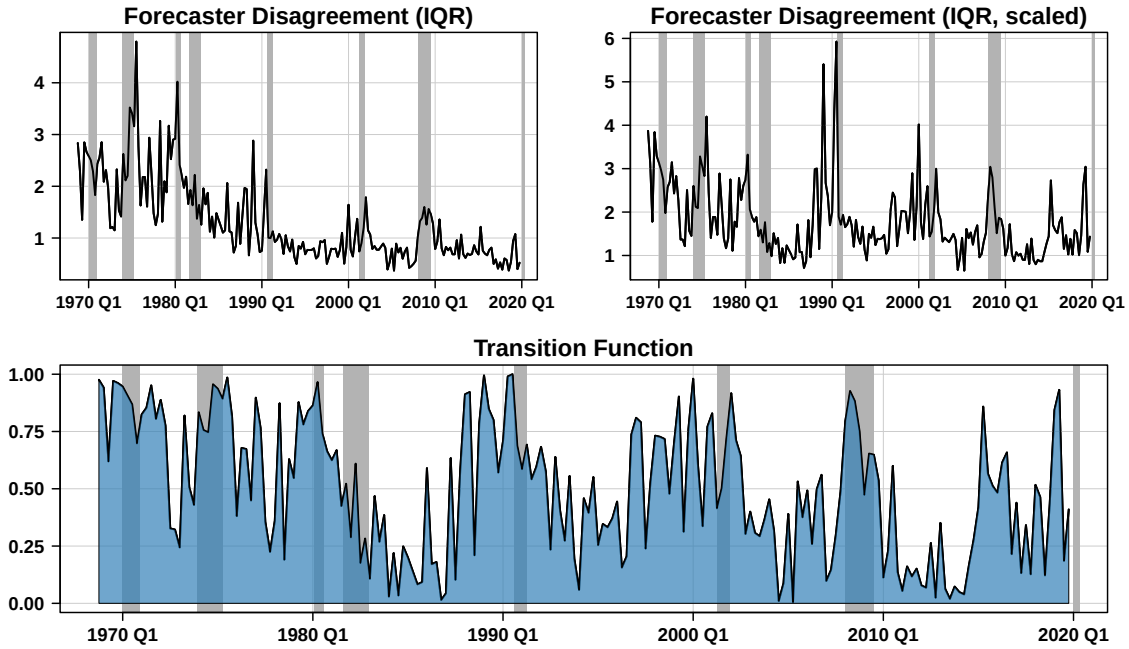
Estimation of Equation (3.5) is done for each horizon and variable separately, resulting in a sequence $\{\beta_h^r\}_{h=1}^H$ that reflects the impulse response for $y_t$ within the first $H$ periods. An important advantage of this approach is that it does not rule out potential regime switches after the shock. The state-dependent local projection framework conditions on being in one of the polar cases before the shock hits but does not make any additional assumptions about the economy staying in a particular regime in subsequent periods (see also the discussion in Ramey and Zubairy, 2018). Rather, the local projection at time $t$ directly provides us with a measure of the conditional average response of an economy in state $z_{t-1}$ going forward. Gonçalves et al. (2024) clarify that LPs recover the conditional average response, given that the state is exogenous.[14] In the case of an endogenous state variable, LPs only recover the conditional marginal response function. We show that disagreement does not react systematically in our case, which lends credibility to the state variable being exogenous to the shock under consideration.

### 3.4 Regime allocation based on low and high disagreement about growth

Before we move on to present the results, we discuss how we construct the transition function that provides us with weights associated with the low-disagreement and high-disagreement regimes. We display our measure of disagreement and the value of the transition function at each point in time in Figure 2. Forecaster disagreement is measured by the interquartile range of forecasters' forecasts in the SPF regarding

[14] Following Gonçalves et al. (2024), if the exogeneity of the state indicator is a problem, the potential bias increases with the size of the shock. They suggest keeping the ratio of the shock magnitude to the standard deviation of the shock small. Hence, we fix $\delta/\sigma_e = 1$.

**Figure 2:** Transition function based on forecaster disagreement.




*Notes*: Top left panel: Interquartile range across forecasters in the SPF regarding output growth in the current quarter, raw series. Top right panel: Interquartile range scaled by the moving average of the standard deviation of output growth (moving average over 24 quarters). Bottom panel: Transition function with weights for being in the high-disagreement regime based on the scaled interquartile range. Gray bars denote the NBER recession dates.

output growth in the current quarter. The raw series is reported in the upper left panel and shows substantial variation in the dispersion of growth forecasts over time. It appears cyclical, with increasing disagreement around most, but not all, of the NBER recessions indicated by the gray areas.

The series, however, is non-stationary. It shows a considerable trend or regime shift in both the level and volatility. In particular, the average level and volatility of dispersion are elevated from the beginning of the sample until the mid-1980s. This pattern is also well documented for inflation disagreement in the SPF, which has been studied more intensively than that for GDP growth. This literature finds forecast disagreement to be positively correlated with the level and volatility of inflation.[15] However, both associations could be purely mechanical if forecast errors were assumed to be proportional to the level or volatility of the forecasted series.

[15]See, for instance, Mankiw et al. (2003) for early evidence that disagreement rises with the level and absolute change in inflation. Coibion and Gorodnichenko (2012) further show that disagreement was persistently higher during periods of volatile inflation, consistent with informational frictions. Dovern et al. (2012) confirm that disagreement is positively associated with both the inflation level and inflation uncertainty across time and countries.

Against this background, we scale the time series of forecaster disagreement by a low-frequency measure of the volatility of GDP growth for our analysis. This accounts for the possibility that a higher variance in GDP growth (and in the underlying shocks) may explain why the dispersion in professional forecasts is higher until the mid-1980s. However, we verify that our results are not driven by the scaling of the series.[16] We compute the standard deviation of GDP growth using a moving window of 24 quarters.[17] We show the adjusted series in the upper right panel of Figure 2. The scaling mutes the large spikes in disagreement in the beginning of the sample but preserves the general pattern of the variable. The highest level of disagreement in the series is now associated with the recessions of the early 1990s and 2000s.

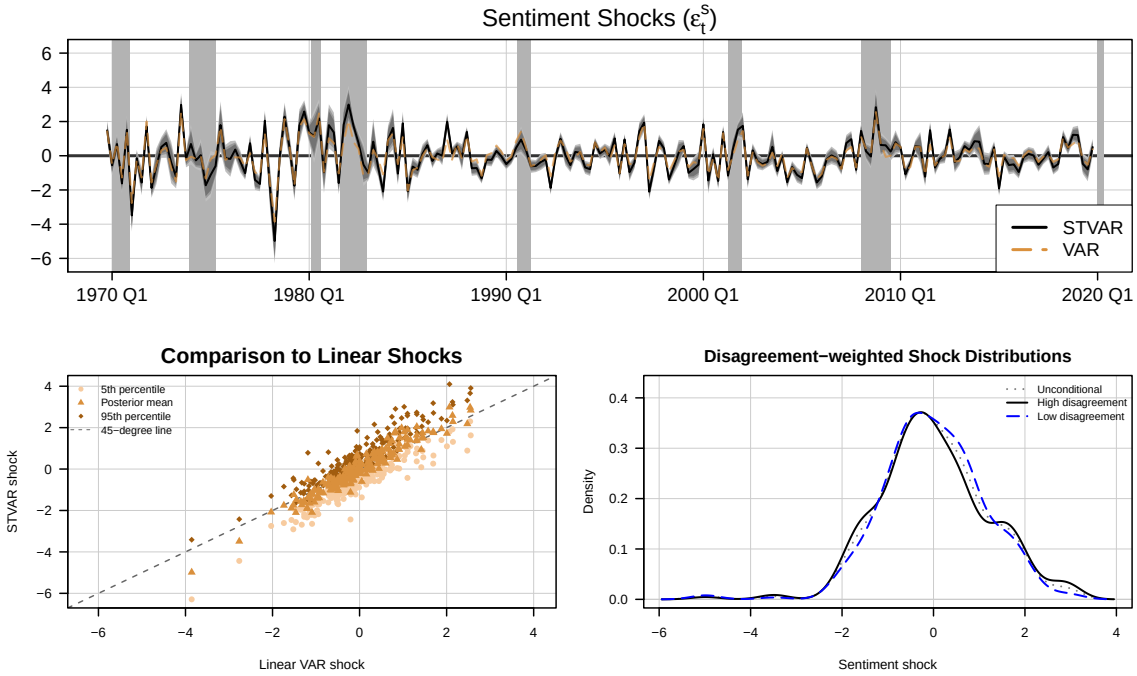
The bottom panel of Figure 2 shows the weight associated with the disagreement regime based on the scaled interquartile range. The values of the transition function are derived using Equation (3.3). We observe that we are hardly ever in one of the two polar regimes of disagreement, with much movement in the medium range. Still, we observe periods of high disagreement throughout the sample and during most of the NBER recessions, with the exception of the Volcker recession. However, non-recessionary periods are also in the high-disagreement regime.

The literature identifies two main sources of disagreement among forecasters: differences in information signals and differences in priors or models. Patton and Timmermann (2010) argue that differences in priors matter primarily for long-run expectations, whereas private information plays a greater role at shorter horizons. Consistent with this view, disagreement about output growth is largest for short-term survey expectations (see, e.g., Coibion and Gorodnichenko, 2012; Coibion and Gorodnichenko, 2015; Andrade et al., 2016). How strongly heterogeneous signals translate into disagreement, however, depends on the information environment. As Section 2 illustrates, fundamental uncertainty and the precision and composition of signals shape both the dispersion of expectations and their sensitivity to noisy information.

[16] Similarly, Falck et al. (2021) observe historically high levels of disagreement in inflation expectations in the 1970s. They scale disagreement by the level of expected inflation to control for periods of relatively high inflation rates.

[17] The empirical results are qualitatively and quantitatively robust to employing alternative window sizes reflecting typical business cycle frequencies of 5 to 7 years. As an alternative approach, we regress disagreement on the standard deviation of GDP growth and retain the residuals. We denote this approach as *purification*; results are also robust to this choice. We discuss these checks in more detail in the robustness section. In an additional robustness exercise, we use the Hodrick-Prescott filter to retrieve the cyclical component of forecasters' disagreement. We report this in Figure B5 in the appendix. Again, results do not change much, neither qualitatively nor quantitatively.

**Figure 3:** Sentiment shocks from the STVAR.




*Notes:* The upper panel reports the posterior mean of the sentiment shocks identified from the STVAR (black solid line), together with the 68%/80%/90% credible sets (gray shaded areas). The orange dashed line reports sentiment shocks identified from a linear VAR. Gray vertical bars indicate NBER recessions. The lower left panel compares the posterior mean and the 5th and 95th percentiles of the STVAR shock estimates (vertical axis) with the linear VAR shock estimates (horizontal axis). The gray dashed line is the 45-degree line. The lower right panel reports kernel density estimates of the posterior mean STVAR shocks, using equal observation weights (gray dotted line), weights proportional to $F(z_{t-1})$ (black solid line), and weights proportional to $1 - F(z_{t-1})$ (blue dashed line). Weights are normalized to sum to one within each density estimate. All three densities use the same bandwidth and evaluation grid. The disagreement weights change the contribution of each observation to the density estimate without rescaling the shock itself.

# 4. Results

This section presents the empirical results. To set the stage, we first contrast the identified shocks with those obtained in a linear framework. We then discuss the main results and conclude with an extensive robustness analysis.

## 4.1 Sentiment shocks: STVAR

We first examine the sentiment shocks, $\varepsilon_t^s$, identified from the STVAR. Their posterior mean is subsequently used to estimate state-dependent responses through local pro-

jections. Recall that identification rests on imposing a negative co-movement between output and the nowcast error on impact in both regimes.

The upper panel of Figure 3 shows substantial variation in the estimated shocks throughout the sample, with several large realizations occurring early in the sample and around recessions. The lower left panel compares the STVAR estimates with those obtained from a linear VAR. The close association holds for both the posterior mean and the reported posterior percentiles, indicating that allowing for state dependence does not substantially change the estimated shock series. This similarity is consistent with imposing the same identifying sign restrictions in both regimes. It does not, however, imply that the subsequent transmission of these shocks is invariant to disagreement.

The lower right panel examines how the distribution of estimated shocks varies with disagreement. We compare the unconditional density with two disagreement-weighted densities: one assigning greater weight to observations with high initial disagreement, using weights proportional to $F(z_{t-1})$, and the other emphasizing observations with low initial disagreement, using weights proportional to $1 - F(z_{t-1})$. The three densities overlap closely, suggesting that the estimated shocks have broadly similar distributions across disagreement states. This comparison is descriptive and does not establish the equality of the underlying distributions.

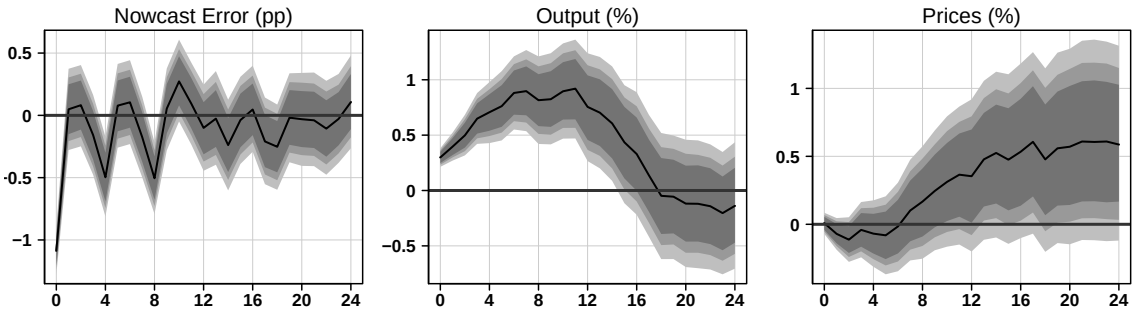
In the appendix, we report the impulse responses of the nowcast error and output to sentiment and non-sentiment shocks obtained directly from the STVAR (see Figure B1). While the overall pattern resembles that documented in earlier studies using linear frameworks (Enders et al., 2021), the output response is stronger in the high-disagreement regime. In what follows, we use STLP to examine these differences in transmission across a broader set of macroeconomic variables.

### 4.2 The effects of sentiment shocks: STLP

To flesh out the transmission mechanism of sentiment shocks under different disagreement regimes, we rely on estimates from STLP. Before presenting the results of the state-dependent model, however, we first show the impulse responses of the nowcast error, output, and prices in Figure 4. We obtain these estimates by suppressing the dependence on the level of disagreement in Eq. (3.5).[18] We display the sequence of $\{\beta_h\}_{h=1}^{H}$, which represents the dynamic causal effects of a sentiment shock. The black solid lines show the median estimates, while the shaded areas indicate the 68%, 80%,

[18] See footnote 12. Note that in the linear specification, the construction of the shock series, $\varepsilon_t^s$, is also done in a linear VAR setting.

**Figure 4:** The linear effects of a sentiment shock.




*Notes*: Estimates based on LP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid lines refer to the median estimate; the gray areas refer to the 68%/80%/90% credible sets. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

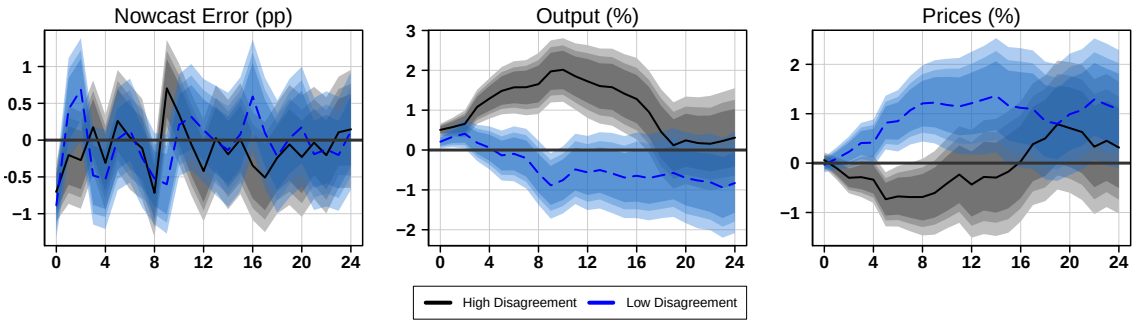
and 90% credible sets. The horizontal axis measures the impulse response horizon in quarters. The vertical axis measures deviations from the trend in percent. In the baseline specification, we include four lags of the dependent variable and the shock series.[19]

The sentiment shock is normalized to unity. This translates into a negative nowcast error of one percentage point; that is, the consensus forecaster is mistakenly excessively *optimistic*: expectations exceed output. The reaction of output is positive—as imposed via the sign restrictions in the impact period to identify a sentiment shock. It is, hence, an expansionary sentiment shock, where optimism about current output growth causes actual output to increase. Prices, however, react only after some time. They increase by a maximum of 0.5 percent compared to the pre-shock level, but this effect is not statistically significant. The positive co-movement of output and prices is consistent with the interpretation of sentiment shocks as demand shocks (Lorenzoni, 2009).

We move on to the presentation of the state-dependent effects of sentiment shocks in Figure 5. We now split the dynamic causal effects of a sentiment shock into two sequences, $\{\beta_h^r\}_{h=1}^H$ for each regime $r \in \{L, H\}$. The black solid lines (gray shaded areas) show the median (68%, 80%, 90% quantiles) estimates for the high

[19]The estimated sentiment shocks, $\varepsilon_t^s$, are generated regressors. Standard errors that ignore the estimation error from the first stage are asymptotically valid under the null hypothesis that the coefficients of interest ($\beta_h^r$) are zero (Pagan, 1984); see also Coibion and Gorodnichenko (2015), footnote 18. In the appendix, we show that the results are robust once we account for the additional first-stage uncertainty; see Figure B3.

**Figure 5:** The state-dependent effects of a sentiment shock.




*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

disagreement case, and the blue dashed line (blue shaded areas) shows the results for the low disagreement case. In the appendix, we report the outcomes once we take the estimation uncertainty of the first stage into account (see Figure B4).

Figure 5 shows the responses of the nowcast error, output, and prices to a sentiment shock normalized, as before, to unity. Again, this means that the consensus forecaster is excessively *optimistic*. The shock transmits differently across regimes. The reaction of the nowcast errors in both regimes is negative on impact, while the reaction of output is positive—the defining feature of a sentiment shock. However, the transmission differs fundamentally depending on the initial level of disagreement. In the case of high disagreement, output increases strongly and persistently. We find a two percent increase in output after two years. The response of prices, in contrast, is basically flat. There is even a mild and short-lived decline 4-6 quarters after the shock takes place, but this is estimated with little statistical precision. We observe the opposite pattern in cases where initial disagreement is low. In this case, prices increase by a full 1.5 percent after two to three years, while output is fairly unresponsive: After the initial increase, it quickly reverts back to its initial level. It even undershoots the pre-shock level; however, this effect is not significant. From around four quarters onward, the credible sets for output and prices separate across regimes. They begin to overlap again after about three years for prices and four years for output.

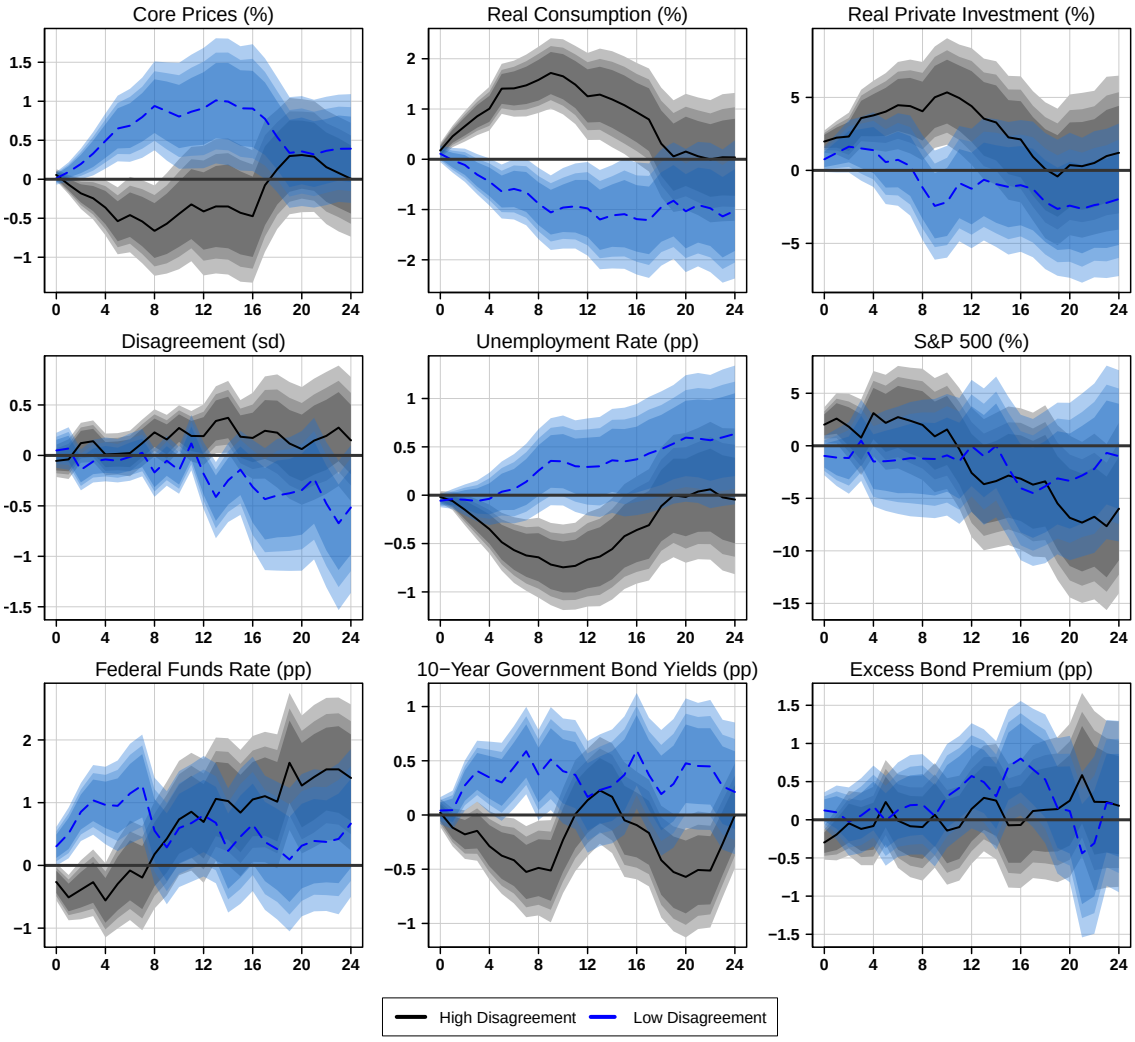
We offer a structural account of these patterns in Section 5. In a nutshell, when aggregate technology is highly volatile—and, hence, disagreement is elevated—agents rely less on their priors. Instead, they place greater weight on the signals they receive, which also makes them more susceptible to noise. Because noise in the public signal reflects sentiment shocks, their positive effects on demand and, subsequently, output become stronger. The muted price response follows from firms' pricing decisions: stronger sentiment shocks lead firms to (incorrectly) expect other firms to set lower prices due to better technology, which—together with strategic complementarities—dampens the price response.

In Figure 6, we show several additional state-dependent effects of a sentiment shock. We report the outcomes of core prices, forecaster disagreement, subcomponents of GDP (consumption and investment), the labor market (unemployment rate), and financial variables (stock market, federal funds rate, long-term rate, and risk premium). Disagreement is standardized to have a unit standard deviation. The impact response is close to zero, with uncertainty bands spanning approximately $-0.2$ to 0.2 standard deviations. We find little evidence that forecaster disagreement, our state variable, responds systematically to the shock.

The shock transmits through both consumption and investment. While differences across regimes are visible for both, these differences are much more pronounced in consumption. The strong transmission through consumption underscores the demand nature of the noise shock. We use the unemployment rate to assess the transmission to the labor market. It mirrors the output response: the unemployment rate decreases in the high-disagreement regime, while it increases slightly, although the increase is not statistically significant, in the low-disagreement regime.

Lastly, we turn to the financial variables. We do not observe strong differences for the S&P 500 stock market index or the excess bond premium as a measure of risk, suggesting that financial factors are unlikely to explain the differences across regimes. For the federal funds rate, we observe different reactions across the regimes. In the low-disagreement regime, the central bank reacts relatively quickly to the rise in prices. In the high-disagreement regime, in line with the negative price response, the central bank takes a (mildly) accommodative stance before turning restrictive after two to three years to prevent the economy from overheating. This pattern is also reflected in longer-term interest rates, as measured by 10-year government bond yields.

**Figure 6:** Additional state-dependent effects of a sentiment shock.




*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the core consumer prices (log), real consumption (log), real private investment (log), forecaster disagreement, the unemployment rate, the S&P 500 (log), the federal funds rate, the 10-year government bond yields, and the excess bond premium. The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percent (core consumer prices, real consumption, real investment, S&P 500), percentage points (unemployment rate, federal funds rate, government bond yield, excess bond premium), or standard deviations (disagreement).

### 4.3 Relation to (technology) news shocks

If the transmission of shocks to expectations depends on disagreement and the underlying information structure, this mechanism should not be specific to sentiment shocks. We therefore examine whether the effects of technology news shocks show a similar state-dependency. In fact, there is a tight connection between noise shocks—or "sentiment shocks"—and news shocks. Chahrour and Jurado (2018) show the observational equivalence between these information structures. Hence, examining technology news shocks provides a complementary test of whether disagreement systematically shapes the transmission of expectation-driven shocks.

Specifically, we examine a news shock to technology that maximizes the forecast error variance of total factor productivity (TFP) at a long but finite horizon (Francis et al., 2014). To isolate the shock, we estimate a state-dependent vector autoregression and use the modified max share approach by Kurmann and Sims (2021) to identify the technology news shock. Then, we use the framework of state-dependent local projections to examine the responses to our core set of variables. The approach is thus equivalent to the one we use for sentiment shocks.

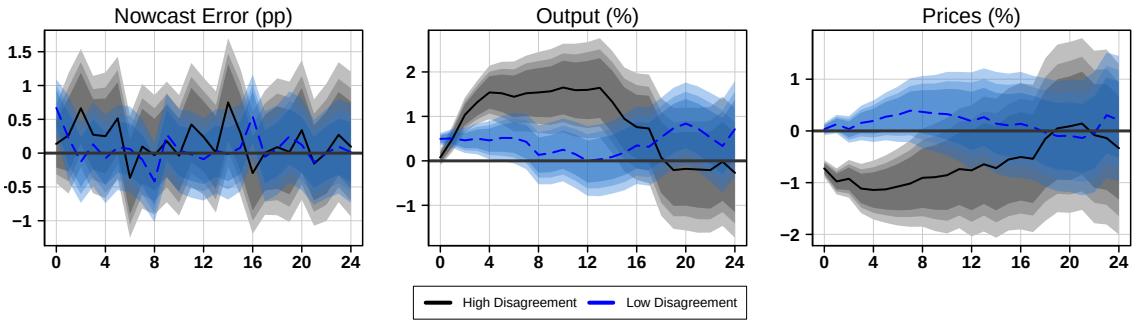
Following the specification of Barsky and Sims (2011), the VAR includes TFP, consumption, output, and hours worked.[20] Consumption and output are measured in real per-capita terms, and hours worked are expressed per capita. All variables enter in log levels. We include four lags and use a Minnesota prior. Identification exploits the assumption that technology news has persistent effects on productivity: the identified shock maximizes the forecast error variance share of TFP at a horizon of 10 years.[21]

We present the results in Figure 7. We show the responses of the nowcast error, output, and prices to a technology news shock normalized to unity. The state-dependent response of output to a news shock is similar to that of a sentiment shock: In the high-disagreement regime, we find stronger output effects than in the low-disagreement regime. We note, however, that the output reaction in the low-disagreement regime is significantly positive for about one year as technology eventually moves. In terms of the nowcast error and prices, the outcomes differ from those of sentiment shocks. We find no strong effect on the nowcast error, reflecting the shock's nature as news about future developments. For prices, we find a negative effect in

[20] We use the TFP measure provided by Fernald (2014), which is based on the growth accounting methodology in Basu et al. (2006) and corrects for unobserved capacity utilization.

[21] We do not impose additional zero-impact restrictions to separate anticipated from surprise shocks to technology (Kurmann and Sims, 2021). This helps to avoid measurement issues that may arise with a variable like TFP in the short run. Our approach is robust to imposing the zero restriction.

**Figure 7:** The state-dependent effects of a news shock.




*Notes*: Estimates based on STLP with an identified technology news shock (Kurmann and Sims, 2021). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

the high-disagreement regime, while in the low-disagreement regime, the outcomes are not statistically significant. The negative effect of technological improvements on prices is consistent with the findings in Barsky and Sims (2011) and Kurmann and Sims (2021).

Taken together, the results show that the state dependence of the real effects is not specific to our identified sentiment shock. Despite being identified using a different setup and reflecting information about future fundamentals rather than noise, the technology news shock generates a similar pattern: Its output effects are stronger when disagreement is high. This finding is consistent with our proposed mechanism whereby disagreement and the underlying information structure shape the transmission of expectation-driven shocks. At the same time, the differences in the responses of nowcast errors and prices highlight that news and sentiment shocks operate through distinct transmission channels.

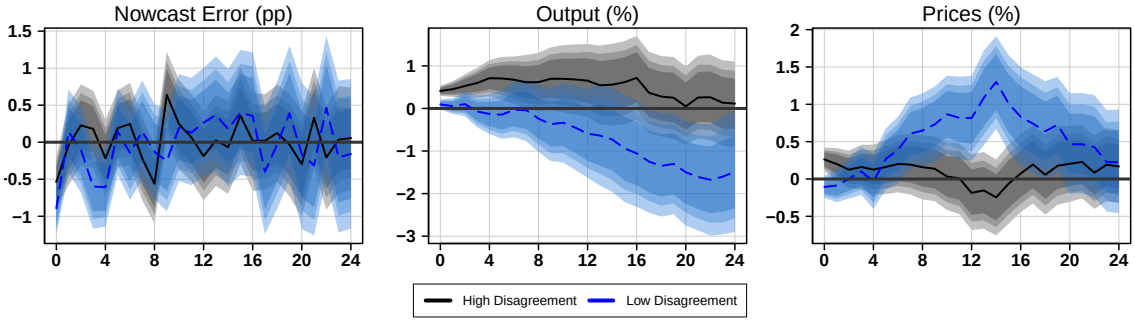
### 4.4 Subsample stability and robustness

We conduct several exercises to inspect the robustness of our main results. As a first check, we re-do the analysis on subsamples to assess the stability of the estimates. The second check involves a series of robustness exercises within the STLP framework. We present the results in Figures 8 and 9.

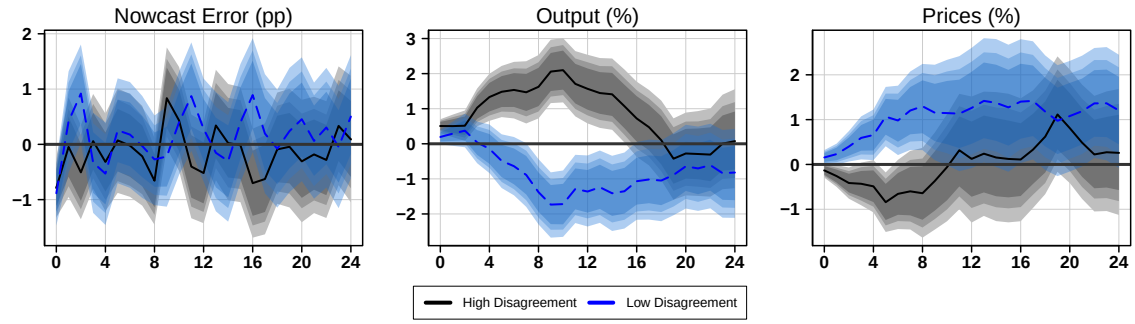
For the subsample stability analysis, we re-estimate both the STVAR and the STLP on shortened subsamples. We investigate two subsamples: a "Post-Volcker

**Figure 8:** Subsample stability of the state-dependent effects of a sentiment shock.

**(a)** Post-Volcker sample: 1983Q1 - 2019Q4




**(b)** Pre-financial-crisis sample: 1969Q4 - 2007Q4

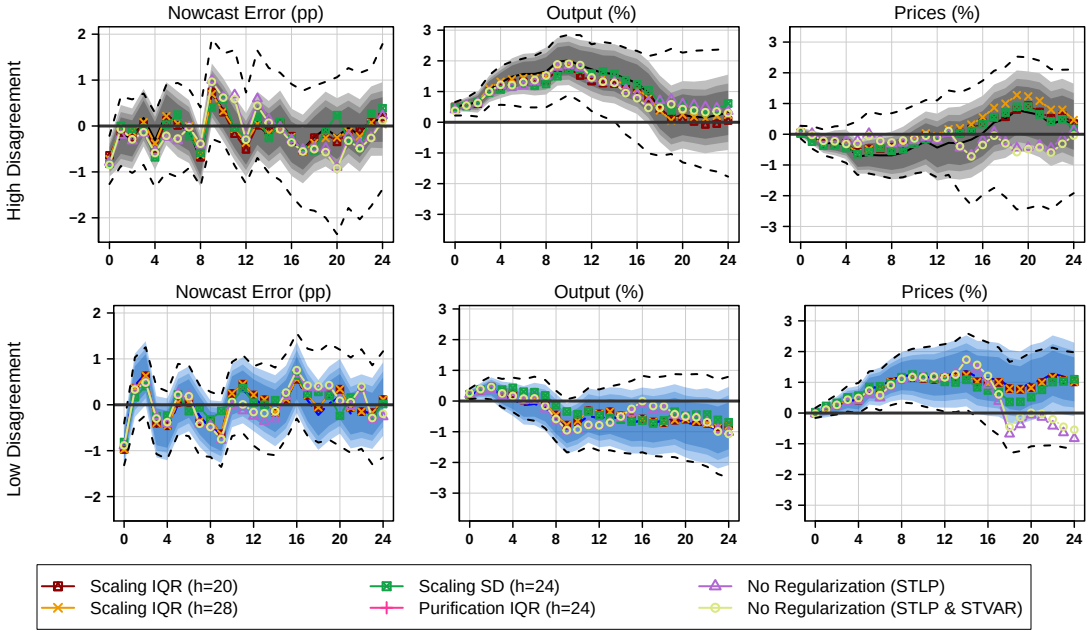



*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers either the period 1983Q1-2019Q4 (upper panel) or 1969Q4-2007Q4 (lower panel). The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

sample" starting in 1983Q1 and a "Pre-financial-crisis sample" ending in 2007Q4. For the former subsample, we show the results in panel (a) of Figure 8. This subsample largely excludes the period of elevated output volatility preceding the Great Moderation (McConnell and Perez-Quiros, 2000).[22] Nevertheless, the results remain similar to those obtained for the baseline sample: In the high-disagreement regime, output reacts more strongly, while the price response is rather muted. On the contrary, in the low-disagreement regime, the output response is muted, and prices increase. In the latter regime, output may even decrease in the long run. Furthermore, magnitudes seem to be subdued in comparison to the baseline results. Next, we turn to the sec-

[22]This check also safeguards against concerns regarding the scaling of disagreement. In the full sample, our results are robust to leaving disagreement unscaled.

**Figure 9:** Robustness to the state-dependent effects of a sentiment shock.




*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. Colored lines with markers refer to robustness specifications: Scaling using a different moving average process (20 or 28 quarters), scaling using the standard deviation as the dispersion measure, using purification instead of scaling, and prior sensitivity analysis (no regularization in the second or both estimation steps). The black dashed lines refer to the maximum/minimum 90% credible interval across all robustness specifications. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

ond subsample reported in panel (b) of Figure 8, in which we exclude all observations starting from the financial crisis. In this case, the impulse responses are similar in shape and magnitude to the baseline results. Again, we observe a decrease in output in the low-disagreement regime.

We also assess the robustness of our results to alternative specifications of the STLP framework in Figure 9.[23] In our baseline, we scale disagreement using the moving-average standard deviation of output growth over the last 24 quarters. The window of 6 years (=24 quarters) reflects a typical business cycle frequency of 5 to 7 years. Changing the window to 5 years (=20 quarters) or 7 years (=28 quarters) does not significantly change the results. We argue that the interquartile range provides

[23] We report robustness checks for the impulse response functions of the remaining variables in the appendix; see Figures B8 and B9.

a more robust measure of disagreement than the standard deviation. Nevertheless, our results remain unchanged when using the standard deviation instead. We also use an alternative approach to transform the disagreement series. Instead of scaling, we regress disagreement on the standard deviation of output growth and retain the residuals. We denote this approach as *purification*; results are again robust to this choice. As we utilize Bayesian shrinkage priors for the regularization of the controls in the empirical framework, we also perform prior sensitivity checks. Results do not change qualitatively or quantitatively when imposing no regularization in the STLP or in both the STLP and the STVAR. In these cases, no regularization means using a completely uninformative prior that resembles ordinary least squares.

We also report the maximum and minimum of the 90% credible interval bounds (period-by-period) across all robustness specifications in the figure. This allows us to examine robustness regarding the second moment. Generally, credible intervals tend to widen when we consider a larger set of specifications. While the main results concerning the nowcast error and output remain unchanged, we observe that the response of prices in the high-disagreement regime turns statistically insignificant. This is consistent with the median price responses across most robustness specifications, which are close to zero. We underscore that we tend to interpret the outcome of prices in the high-disagreement regime as a null effect.

We provide additional robustness checks in Appendix B. First, Figure B5 provides robustness regarding the number of lags and additional control variables. Second, Figure B6 cross-validates the state-dependency with indicators for uncertainty as the state variable. Instead of disagreement, we use the financial and macroeconomic uncertainty measures provided by Jurado et al. (2015). Furthermore, we use an alternative version of the disagreement series, scaled by the real-time moving standard deviation of output growth. Third, we control for the state of the business cycle to rule out an alternative interpretation whereby our results are driven not by information frictions but by differences in capacity utilization over the business cycle. In Figure B7, we show that our results still hold in this case. Fourth, the main response patterns remain robust when we control for SPF forecast revisions and identified technology news shocks; see Figure B10. Finally, we assess informational sufficiency by using a shock-predictability test following Forni and Gambetti (2014). We test whether lagged information from a high-dimensional macroeconomic dataset (FRED-QD, McCracken and Ng, 2020) predicts the identified sentiment shocks. The test detects predictability in the baseline shocks. Expanding the information set

in the STVAR eliminates statistically detectable predictability while preserving the main state-dependent response patterns; see Appendix B.3.

## 5. Theory

To provide a structural interpretation of the evidence, we develop a model that embeds the signal-extraction problem sketched in Section 2 in a microfounded New Keynesian framework. The model builds on the noisy and dispersed information framework of Lorenzoni (2009), which we simplify—most notably by assuming predetermined rather than staggered prices—so that the resulting model admits a closed-form solution. This setup allows us to study the macroeconomic effects of sentiment shocks in general equilibrium. We first show that the informational friction generates nowcast errors consistent with the sign restrictions imposed above. Most importantly, we show that uncertainty amplifies the impact of sentiment shocks.

### 5.1 Model outline

In this section, we provide a compact description of the model; see Lorenzoni (2009) for further details of the original model.

**Setup and timing.** There is a continuum of islands (or locations), indexed by $l \in [0, 1]$, each populated by a representative household and a unit mass of producers, indexed by $j \in [0, 1]$. Each household buys from a subset of all islands, chosen randomly in each period. Specifically, it buys from all producers on $n$ islands included in the set $\mathcal{B}_{l,t}$, with $1 < n < \infty$.[24] Households have an infinite planning horizon. Producers produce differentiated goods on the basis of island-specific productivity, which is determined by a permanent, economy-wide component and a temporary, idiosyncratic component. Both components are stochastic.[25] Financial markets are complete such that, assuming identical initial positions, wealth levels of households are equalized at the beginning of each period.

The timing of events is as follows: each period consists of three stages. During stage one of period $t$, information about all variables of period $t-1$ is released.

[24] This setup ensures that households cannot exactly infer aggregate productivity from observed prices. At the same time, individual producers have no impact on the price of households' consumption baskets.

[25] As argued by Lorenzoni (2009), these assumptions can account for the empirical observations that the firm-level volatility of productivity is large relative to aggregate volatility and that individual expectations are dispersed.

Subsequently, nominal wages are determined, and the central bank sets the interest rate based on expected inflation. Shocks emerge during the second stage. We distinguish between shocks that are directly observable and shocks that are not. Sentiment and technology shocks are not directly observable in the following sense: information about idiosyncratic productivity is private to each producer, but, in addition, all agents observe a signal about average productivity. While the signal is unbiased, it contains an i.i.d. zero-mean component: the sentiment shock. We additionally allow for one generic shock that is observable. To simplify the discussion, we refer to this shock as a "monetary policy shock," with the understanding that other observable shocks would play a comparable role in terms of identification. Given this information set, producers set prices.

During the third and final stage, households split up. Workers work for all firms on their island, while consumers allocate their expenditures across differentiated goods based on private and public information, including the signals and the information contained in the prices of the goods in their consumption bundle. Because the common productivity component is permanent and households' wealth and information are equalized in the next period, agents expect the economy to settle on a new steady state from period $t+1$ onward.

**Households.** A representative household on island $l$ ("household $l$", for short) maximizes lifetime utility, given by

$$U_{l,t} = \mathbb{E}_{l,t} \sum_{k=t}^{\infty} \beta^{k-t} \ln C_{l,k} - \frac{L_{l,k}^{1+\varphi}}{1+\varphi} \qquad \varphi \geq 0, \quad 0 < \beta < 1,$$

where $\mathbb{E}_{l,t}$ is the expectation operator based on household $l$'s information set at the time of its consumption decision in stage three of period $t$ (see below). $C_{l,t}$ denotes the consumption basket of household $l$, while $L_{l,t}$ is its labor supply. The flow budget constraint is given by

$$\mathbb{E}_t \varrho_{l,t,t+1} \Theta_{l,t} + B_{l,t} + \sum_{m \in \mathcal{B}_{l,t}} \int_0^1 P_{j,m,l,t} C_{j,m,l,t} dj \leq$$
$$\int_0^1 \Pi_{j,l,t} dj + W_{l,t} L_{l,t} + \Theta_{l,t-1} + (1 + r_{t-1}) B_{l,t-1},$$

where $C_{j,m,l,t}$ denotes the amount bought by household $l$ from producer $j$ on island $m$, and $P_{j,m,l,t}$ is the price for one unit of $C_{j,m,l,t}$. At the beginning of the period, household $l$ receives the payoff $\Theta_{l,t-1}$, given a portfolio of state-contingent securities purchased in the previous period. $\Pi_{j,l,t}$ are the profits of firm $j$ on island $l$, and $\varrho_{l,t,t+1}$

is household $l$'s stochastic discount factor between $t$ and $t+1$. The period-$t$ portfolio is priced conditionally on the (common) information set of stage one. We therefore apply the expectation operator $\mathbb{E}_t$. $B_{l,t}$ are state-non-contingent bonds paying an interest rate of $r_t$. The complete set of state-contingent securities is traded in the first stage of the period, while state-non-contingent bonds can be traded via the central bank throughout the entire period. The interest rate of the non-contingent bond is set by the central bank. All financial assets are in zero net supply. The bundle $C_{l,t}$ of goods purchased by household $l$ consists of goods sold in a subset of all islands,

$$C_{l,t} = \left( \frac{1}{n} \sum_{m \in \mathcal{B}_{l,t}} \int_0^1 C_{j,m,l,t}^{\frac{\gamma-1}{\gamma}} dj \right)^{\frac{\gamma}{\gamma-1}} \qquad \gamma > 1.$$

While each household purchases a different random set of goods, we assume that the number $n$ of islands visited is the same for all households. The price index of household $l$ is therefore

$$P_{l,t} = \left( \frac{1}{n} \sum_{m \in \mathcal{B}_{l,t}} \int_0^1 P_{j,m,l,t}^{1-\gamma} dj \right)^{\frac{1}{1-\gamma}}.$$

**Producers and monetary policy.** The central bank follows an interest-rate feedback rule but sets $r_t$ before observing prices, that is, during stage one of period $t$:

$$r_t = \psi \mathbb{E}_{cb,t} \pi_t + \nu_t \qquad \psi > 1,$$

where $\pi_t$ is economy-wide net inflation, calculated based on all goods sold in the economy. The expectation operator $\mathbb{E}_{cb,t}$ is conditional on the information set of the central bank. This set consists of information from period $t$–1 only, that is, the central bank enjoys no informational advantage over the private sector.[26] $\nu_t$ is a monetary policy shock that is observable by producers and households alike.

Producer $j$ on island $l$ produces according to the following production function

$$Y_{j,l,t} = A_{j,l,t} L_{j,l,t}^{\alpha} \qquad 0 < \alpha < 1,$$

featuring labor supplied by the local household as the sole input. $A_{j,l,t} = A_{l,t}$ denotes the productivity level of producer $j$, which is the same for all producers on island $l$. During stage two, the producer sets her optimal price for the current period. Given prices, the level of production is determined by demand during stage three.

[26] Pre-set prices and interest rates allow us to discard the noisy signals about quantities and inflation observed by producers and the central bank in Lorenzoni (2009), simplifying the signal-extraction problem without changing the qualitative predictions of the model. Pre-set wages, on the other hand, guarantee price-level determinacy. They do not affect output dynamics after sentiment and technology shocks, because goods prices may still adjust in the second stage of the period.

**Sentiment shocks.** Log-productivity on each island is the sum of an aggregate and an island-specific idiosyncratic component

$$a_{l,t} = x_t + \eta_{l,t},$$

where $\eta_{l,t}$ is an i.i.d. shock with variance $\sigma_\eta^2$ and mean zero. This variance causes the dispersion of expectations. A higher value of $\sigma_\eta^2$ also worsens the signal-to-noise ratio of the signals received by private agents, which can create a positive link between dispersion and uncertainty depending on the initial precision of private information. The idiosyncratic shock aggregates to zero across all islands. The aggregate component $x_t$ follows a random walk

$$\Delta x_t = \varepsilon_t.$$

The i.i.d. productivity shock $\varepsilon_t$ has a variance $\sigma_\varepsilon^2$ and a mean of zero. During stage two of each period, agents observe a public signal about $x_t$. This signal takes the form

$$s_t = \varepsilon_t + e_t,$$

where $e_t$ is an i.i.d. sentiment shock with variance $\sigma_e^2$ and mean zero. Note that, since $a_{l,t} - x_{t-1} = \varepsilon_t + \eta_{l,t}$, the signal-extraction problem is identical to that in Section 2. Hence, producers' expectations of $\Delta x_t$ are

$$\mathbb{E}_{j,l,t}\Delta x_t = \rho_x^p s_t + \delta_x^p (a_{j,l,t} - x_{t-1}),$$

with $\mathbb{E}_{j,l,t}$ being the expectation of producer $j$ on island $l$ when setting prices (in stage two). The coefficients $\rho_x^p$ and $\delta_x^p$ are the same for all producers, where these and the following $\rho$ and $\delta$-coefficients are functions of the structural parameters that capture the informational friction. They are non-negative and smaller than unity; see Appendix D. Finally, while shopping during stage three, consumers observe a set of prices. Given that they have also observed the signal, they can infer the productivity level of each producer in their sample. Consumers' expectations are thus given by

$$\mathbb{E}_{l,t}\Delta x_t = \rho_x^h s_t + \delta_x^h \tilde{a}_{l,t},$$

where $\tilde{a}_{l,t}$ is the average over the realizations of $a_{m,t} - x_{t-1}$ for each island $m$ in household $l$'s sample. $\rho_x^h$ and $\delta_x^h$ are equal across households. The model nests the case of complete information about all relevant variables for households and producers if $\sigma_e^2 = 0$. If $\sigma_e^2 > 0$, producers will set prices based on potentially overly optimistic or pessimistic expectations of productivity. Consumers also have complete information if $n \to \infty$.

**Market clearing.** Goods and labor markets clear in each period:

$$\int_0^1 C_{j,m,l,t} dl = Y_{j,m,t} \ \ \forall j, m \qquad L_{l,t} = \int_0^1 L_{j,l,t} dj \ \ \forall l,$$

where $C_{j,m,l,t} = 0$ if household $l$ does not visit island $m$. The asset market clears in accordance with Walras' law.

### 5.2 Results

We now present the results obtained from a linear approximation of the equilibrium conditions around the symmetric steady state; see Appendix D for details. Lowercase letters denote percentage deviations from the steady state. Our first result concerns which shocks generate nowcast errors and their co-movement with output; the second establishes the state dependence—how uncertainty shapes the impact of sentiment shocks. Proofs are provided in Appendix E.

Our first result provides the underpinning for our identification strategy.

**Proposition 1** *A sentiment shock, $e_t$, induces a negative correlation between the reactions of output and the nowcast error, while a technology shock, $\varepsilon_t$, induces a positive correlation. A monetary policy shock, $\nu_t$, does not cause a nowcast error. Formally, we have*

$$y_t = x_{t-1} + \underbrace{\rho_x^h (1-\Omega)}_{>0} e_t + \underbrace{\left[(\delta_x^h + \rho_x^h)(1-\Omega) + \Omega\right]}_{>0} \varepsilon_t \underbrace{- \frac{\alpha}{\alpha + \psi(1-\alpha)}}_{<0} \nu_t, \tag{5.6}$$

*with* $0 < \Omega = \frac{n - \delta_x^h (1-\alpha)[(n-1)\delta_x^p + 1]}{n\alpha + (1-\alpha)\left\{(1-\delta_x^h)[1 + \delta_x^p (n-1)] + (n-1)\gamma(1-\delta_x^p)\right\}} < 1$, *and*

$$y_t - \mathbb{E}_{k,t} y_t = \underbrace{-\rho_x^k \left[\delta_x^h (1-\Omega) + \Omega\right]}_{<0} e_t + \underbrace{\left[\delta_x^h (1-\Omega) + \Omega\right](1 - \delta_x^k - \rho_x^k)}_{>0} \varepsilon_t, \tag{5.7}$$

*with $\mathbb{E}_{k,t}$ standing for either the expectations of producers, $\mathbb{E}_{j,l,t}$, or households, $\mathbb{E}_{l,t}$, and $\rho_x^k, \delta_x^k$ correspondingly for $\rho_x^p, \delta_x^p$ or $\rho_x^h, \delta_x^h$.*

Hence, positive productivity and sentiment shocks raise actual output but also lead to output misperceptions. Consider first a positive sentiment shock, which leaves actual aggregate technology unchanged. Producers expect aggregate productivity to be high—resulting in higher demand—but also observe that their own productivity is unchanged, which they attribute to a negative realization of the idiosyncratic productivity component. Consequently, they set prices above the average price they expect other producers to charge. Consumers, in turn, observe prices that are higher than

expected given the public signal. They attribute this to unfavorable idiosyncratic productivity among the firms in their sample and anticipate lower prices in the future, creating an incentive to postpone consumption. At the same time, perceived permanently higher technology raises expected income and current demand. This income effect dominates the substitution effect, which is limited because prices already fall on impact. Current expenditure and output therefore rise.[27] Yet, because producers and consumers both perceive themselves as relatively unlucky, they overestimate current aggregate output, generating a negative nowcast error.

After a positive productivity shock, producers do not fully trust the signal about the aggregate component and attribute some of the increased productivity to an idiosyncratic advantage. They, therefore, reduce prices below what they expect the average price level to be. Consumers, in turn, observe lower prices and expect higher income. They consequently raise consumption. However, both producers and their customers expect other producers to set higher prices and consequently underestimate aggregate output. This generates a positive nowcast error, inducing an opposite correlation between output and the nowcast error for sentiment and productivity shocks.

Finally, monetary policy shocks do not generate nowcast errors because they are fully observed. More generally, only imperfectly observed shocks, such as innovations in productivity or sentiment shocks, give rise to misperceptions about economic activity.

Our second result establishes the state dependence—how uncertainty shapes the impact of sentiment shocks.

**Proposition 2** *An increase in fundamental uncertainty, $\sigma_\varepsilon^2$, increases the dispersion of output nowcasts and forecasts among firms as well as households and reduces the price response to positive sentiment shocks. It also amplifies the output response to positive sentiment shocks if*

$$Q\left(\frac{\sigma_\eta^2/n}{\sigma_\varepsilon^2}+\frac{\sigma_\eta^2/n}{\sigma_e^2}\right)^2>1, \tag{5.8}$$

*where $Q>1$ is a constant defined in Appendix E.*

[27] As pointed out by Lorenzoni (2009), the sentiment shock provides a possible microfoundation for the traditional concept of a demand shock: agents are too optimistic about economic fundamentals, resulting in unusually high demand.

Proposition 2 thus shows how the model can account for our empirical findings. To build intuition, recall from Section 2 that greater fundamental uncertainty shifts weight from priors to signals. Placing more weight on the public signal leads firms to expect lower prices from competitors and, given strategic complementarities in price setting, to lower their own prices further. Households also rely more on their signals, strengthening both the perceived income gain and the incentive to postpone consumption described above. The output response increases if the stronger income effect outweighs the stronger substitution effect, that is, if Condition (5.8) holds.

We argue that this condition is likely satisfied for empirically plausible parameter values. To see why, consider the exceptional case in which households' private observations are very precise relative to both fundamental uncertainty ($\sigma_\eta^2/n \ll \sigma_\varepsilon^2$) and noise in the public signal ($\sigma_\eta^2/n \ll \sigma_e^2$). Remember that $\sigma_\eta^2/n$ is the effective noise in households' private signal, the average of observed prices.

The intuition is as follows. Higher technology volatility increases households' weights on both signals. When households' private observations are very precise, however, the public signal receives little additional weight relative to those observations. Higher technology volatility therefore raises perceived permanent income only slightly. At the same time, firms place more weight on their own productivity as evidence about aggregate technology, making prices more responsive to actual technology, relative to the public signal. Households who mistake part of a sentiment shock for a technology improvement consequently expect prices to fall further relative to what they currently pay. Although this expected further decline may itself be small, its increase can outweigh the even smaller increase in perceived income. In that case, the additional substitution effect dominates the additional income effect, and the output response to a sentiment shock declines as technology volatility rises.

Empirically, however, this exceptional case seems unlikely. Idiosyncratic technology shocks tend to be much more volatile than aggregate shocks (see, e.g., Comin and Philippon, 2005, Franco and Philippon, 2007, Lorenzoni, 2009, Andrade et al., 2022, or Dovern et al., 2023), and households are unlikely to have a very large informational advantage over firms, so that $n$ is unlikely to be very large. Moreover, the condition holds if the public signal is at least as informative as households' private signal, implying $\sigma_e^2 \leq \sigma_\eta^2/n$. This holds if, for instance, an agency/media outlet that produces the public signal collects at least as much information as an average household.

In sum, under plausible informational assumptions, higher fundamental uncertainty generates greater disagreement and stronger real effects of sentiment shocks, while making the price response more negative.

## 6. Conclusion

This paper studies the conditions under which the economy becomes susceptible to non-fundamental shocks. We focus on expectation-driven business cycle disturbances, which we label "sentiment shocks." Our main finding is that their macroeconomic effects depend critically on the degree of disagreement among forecasters. When disagreement is high, sentiment shocks have sizable effects on economic activity while leaving prices largely unaffected. When disagreement is low, their effects are mostly absorbed by prices and have little impact on output.

To interpret these findings, we rely on a New Keynesian model with dispersed information in which agents use noisy signals to infer aggregate productivity. We show that when uncertainty about fundamentals rises, agents place greater weight on these signals, amplifying the real effects of sentiment shocks. In such environments, agents must infer fundamentals from noisy signals—dancing in the dark.

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# Online Appendix: Dancing in the Dark: Sentiment Shocks and Economic Activity

## A. Further details on data and econometric methodology

### A.1 Data sources

All series were gathered from the sources listed below, which include the Bureau of Economic Analysis (BEA), the Federal Reserve Economic Data (FRED) database, the Real-Time Data Set for Macroeconomists, and the Survey of Professional Forecasters (SPF) provided by the Federal Reserve Bank of Philadelphia. In Table A1, we provide an overview of the data, their transformations, and sources.

**Table A1:** Data definitions, transformations, and sources.

| Variable ($y_{it}$) | Transformation | Mnemonic | Source |
|---|---|---|---|
| Output | $100 \log x_t$ | `GDPC1` | FRED |
| Output (first-release) | $x_t$ | `routput` | BEA |
| Output Nowcast | $y_t$ | `RGDP` | SPF |
| Output Disagreement | $D(x_t)$ | `RGDP` | SPF |
| Consumer prices | $100 \log x_t$ | `CPIAUCSL` | FRED |
| Core prices | $100 \log x_t$ | `CPILFESL` | FRED |
| GDP deflator | $100 \log x_t$ | `GDPDEF` | FRED |
| Real consumption | $100 \log x_t$ | `PCE` | FRED |
| Real private investment | $100 \log x_t$ | `GPDI` | FRED |
| Unemployment rate | $x_t$ | `UNRATE` | FRED |
| Federal funds rate | $x_t$ | `FEDFUNDS` | FRED |
| 10-year government bond yields | $x_t$ | `GS10` | FRED |
| Excess bond premium | $x_t$ | `EBP` | Favara et al. (2016) |
| S&P 500 | $100 \log x_t$ | `P` | Robert Shiller's website |
| Financial uncertainty ($h = 1$) | $x_t$ | `finunc` | Sydney Ludvigson's website |
| Macro uncertainty ($h = 1$) | $x_t$ | `macrounc` | Sydney Ludvigson's website |

*Notes:* The dispersion function $D(y_{it})$ refers to either the standard deviation or the interquartile range. Real private consumption and real investment are deflated by the GDP deflator.

### A.2 Smooth-transition vector autoregression

As detailed in Section 3, we use a bivariate smooth-transition vector autoregression (STVAR) to identify sentiment shocks non-linearly. In this appendix, we provide additional information on the model and its estimation. The time series process $\{\boldsymbol{y}_t\}_{t=1}^{T}$ $(2 \times 1)$ follows

$$\begin{aligned} \boldsymbol{y}_t = &(\boldsymbol{c}_{11} + \boldsymbol{A}_{11}\boldsymbol{y}_{t-1} + \ldots + \boldsymbol{A}_{1p}\boldsymbol{y}_{t-p}) \times F(z_{t-1}) \\ &+ (\boldsymbol{c}_{21} + \boldsymbol{A}_{21}\boldsymbol{y}_{t-1} + \ldots + \boldsymbol{A}_{2p}\boldsymbol{y}_{t-p}) \times (1 - F(z_{t-1})) \\ &+ \boldsymbol{c}_2 t + \boldsymbol{c}_3 t^2 + \boldsymbol{u}_t, \qquad\qquad \boldsymbol{u}_t \sim \mathcal{N}(0, \boldsymbol{\Sigma_t}), \end{aligned} \tag{A.1}$$

where $\boldsymbol{A}_{rj}$ are $2 \times 2$ coefficient matrices for regime $r \in \{1, 2\}$ and lag $j \in \{1, 2, \ldots, p\}$, the $2 \times 1$ vectors $\boldsymbol{c}_{r1}$, $\boldsymbol{c}_2$, and $\boldsymbol{c}_3$ denote the coefficients corresponding to the intercept, trend, and quadratic trend. The $2 \times 1$ vector $\boldsymbol{u}_t$ denotes the reduced-form innovations, which are normally distributed with a zero mean and time-varying variance $\boldsymbol{\Sigma}_t$. The time variation results from the state dependency as follows

$$\boldsymbol{\Sigma}_t = \tilde{\boldsymbol{\Sigma}}_1 F(z_{t-1}) + \tilde{\boldsymbol{\Sigma}}_2 (1 - F(z_{t-1})), \tag{A.2}$$

where $F(z_{t-1})$ denotes the transition function. It reads as follows

$$F\left(z_{t-1}\right) = \frac{1}{T} \sum_{j=1}^{T} \mathbb{1}(z_j \leq z_{t-1}), \tag{A.3}$$

where $T$ is the number of observations in our sample, $\mathbb{1}$ is an indicator function, and $j$ indexes all observations. The transition weight equals one at the sample maximum and $1/T$ at the sample minimum in the absence of ties. Values approaching zero and one represent the limiting low-disagreement and high-disagreement cases.

As an alternative, we also use the logistic function as a transition function, as is often done in the literature (Auerbach and Gorodnichenko, 2012; Caggiano et al., 2014; Tenreyro and Thwaites, 2016; Falck et al., 2021). Therefore, we specify the transition function based on the logistic function, following Granger and Teräsvirta (1993). We assume that the alternative $\tilde{F}(z_{t-1})$ follows

$$\tilde{F}\left(z_{t-1}\right) = \frac{\exp\left(\theta \frac{z_{t-1} - c}{\sigma_z}\right)}{1 + \exp\left(\theta \frac{z_{t-1} - c}{\sigma_z}\right)}, \tag{A.4}$$

where $c$ corresponds to the mean and $\sigma_z$ to the standard deviation of $z_{t-1}$. The smoothness parameter $\theta$ determines the curvature of $\tilde{F}(z_{t-1})$ and how strongly the economy switches from the low-disagreement to the high-disagreement regime when $z_t$ changes. Several previous studies have calibrated, rather than estimated, the smooth-

ness parameter $\theta$. We follow their suggestion and use a value of $\theta = 5$. Results are reported in Figure B2.

The model is identified using sign restrictions. We impose our set of sign restrictions using the methods in Rubio-Ramirez et al. (2010). The reduced-form innovations, $\boldsymbol{u}_t$, are related to the structural innovations, $\boldsymbol{\varepsilon}_t$, by the linear combination $\boldsymbol{u}_t = \boldsymbol{S}_t\boldsymbol{\varepsilon}_t$, where structural shocks are, by assumption, orthogonalized such that $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ and $\boldsymbol{S}_t$ is a $2 \times 2$ nonsingular matrix. The time variation in the impact matrix $\boldsymbol{S}_t$ arises from the time variation in the covariance matrix, as $\boldsymbol{\Sigma}_t = \boldsymbol{S}_t\boldsymbol{S}_t'$. Given that this time variation is exclusively driven by two states, we obtain $\tilde{\boldsymbol{\Sigma}}_r = \tilde{\boldsymbol{S}}_r\tilde{\boldsymbol{S}}_r'$, where $\boldsymbol{S}_r$ is the regime-specific structural impact matrix with $r \in \{1, 2\}$. To implement the sign restrictions, we proceed as follows. The regime-specific covariance matrix can be rewritten as $\tilde{\boldsymbol{\Sigma}}_r = \boldsymbol{C}_r\boldsymbol{Q}\boldsymbol{Q}'\boldsymbol{C}_r'$, where $\boldsymbol{C}_r$ denotes the lower Cholesky factor of $\boldsymbol{\Sigma}$ and $\boldsymbol{Q} \in \mathcal{O}(2)$, which is the set of all $2\times2$ orthonormal matrices ($\boldsymbol{Q}\boldsymbol{Q}' = \boldsymbol{I}$ holds). We use the algorithm of Rubio-Ramirez et al. (2010) to independently draw from the uniform distribution over $\mathcal{O}(2)$. We search for a $\boldsymbol{Q}$ such that the sign restrictions on the structural impact matrix $\tilde{\boldsymbol{S}}_r = \boldsymbol{C}_r\boldsymbol{Q}$ are satisfied. This methodology takes into account both the uncertainty inherent to sign restrictions and the uncertainty of the estimated parameters.

The model estimation is done in a Bayesian fashion, and thus we discuss our prior choices. Conditional on the state indicator function $F(z_{t-1})$, the model is linear. Hence, we collect VAR parameters of regime $r$ in the $K \times 1$ vector $\boldsymbol{a}_r = \text{vec}(\boldsymbol{A}_{r1}, \ldots, \boldsymbol{A}_{rp})$ with $K = n^2p$. The prior variances are collected in $\underline{\boldsymbol{V}_r} = \text{diag}(\underline{v_{r1}}, \ldots, \underline{v_{rK}})$, with $\underline{v_k} = \lambda_r^2\psi_{rk}^2$ for $k = 1, \ldots, K$ and $r \in \{1, 2\}$. We propose a hierarchical global-local shrinkage prior setup based on the horseshoe (HS) prior following Carvalho et al. (2010), which reads as follows

$$\boldsymbol{a}_r \sim \mathcal{N}_k\left(\underline{\boldsymbol{a}_r}, \underline{\boldsymbol{V}_r}\right), \quad \lambda_r \sim C^+(0, 1), \quad \psi_{rk} \sim C^+(0, 1), \tag{A.5}$$

where $C^+$ denotes the half-Cauchy distribution. The parameter $\lambda_r$ denotes the global shrinkage parameter of regime $r$, which exerts shrinkage on all coefficients. The parameter $\psi_{rk}$ allows for individual, coefficient-specific shrinkage. Both parameters are estimated, and we do not have to specify any hyperparameters. We use the procedure outlined in Makalic and Schmidt (2015) to sample from the corresponding conditional posterior densities. We center the prior distribution of $\boldsymbol{a}_r$ on $\underline{\boldsymbol{a}_r}$, which is either unity for series in log-levels (mimicking the Minnesota-type prior in Doan et al., 1984, and Litterman, 1986) or zero in all other cases. For the deterministic terms,

we assume a non-informative Gaussian prior $\boldsymbol{c}_l \sim \mathcal{N}(0, 10^5)$ where $l = \{0, 1, 2\}$. The prior distribution of the covariance matrix is

$$\tilde{\boldsymbol{\Sigma}}_r \sim iW(\underline{\nu}, \underline{\boldsymbol{S}}), \tag{A.6}$$

where $iW(\nu, \boldsymbol{S})$ denotes the inverse Wishart distribution with prior degrees of freedom $\nu$ and prior scaling matrix $\boldsymbol{S}$. Following the recommendation in Kadiyala and Karlsson (1997), we specify $\underline{\nu} = n + 2$ and $\underline{\boldsymbol{S}} = \text{diag}(s_1,^2, \ldots, s_n^2)$. Here, the diagonal elements of the scaling matrix $s_j^2$ $(j = 1, \ldots, n)$ denote the sample variance of the residuals of an AR(4) process for each individual series. If we also estimate the parameter of the transition function, we must specify a prior density. For the speed of adjustment parameter $\theta$, we then specify a Gamma distribution such that

$$\theta \sim \mathcal{G}(\underline{a}, \underline{b}), \tag{A.7}$$

where $\underline{a} = 20$ and $\underline{b} = 4$. This translates into a prior mean of 5 and a prior variance of 1.25. In case we do not estimate this parameter, we fix it to 5.

As a last step, we briefly discuss the posterior simulation within an MCMC algorithm. Conditional on the regime indicator function $F(z_{t-1})$, the model is fully linear, and all conditional posterior distributions are available in closed form, rendering a Gibbs sampler convenient. Although the conditional posterior distributions of the VAR coefficients and the covariance matrix are standard, we use the auxiliary sampler of Makalic and Schmidt (2015) to sample from the conditional posterior densities for the HS prior. Lastly, if necessary, the parameter related to the transition function, $\theta$, must be sampled with the help of a Metropolis-Hastings within-Gibbs step since the posterior distribution is not available in closed form.

### A.3 State-dependent Bayesian local projections

We summarize the state-dependent local projections (LP) specified in Equation (3.5) in the following formulation

$$\Delta^h y_{t+h} = \boldsymbol{\varepsilon}_t^{'} \boldsymbol{\beta}_h + \boldsymbol{Z}_{t-1} \boldsymbol{\gamma}_h + u_{t+h}^{(h)}, \qquad u_{t+h}^{(h)} \sim \mathcal{N}(0, \sigma_h^2), \tag{A.8}$$

where $\Delta^h = y_{t+h} - y_{t-1}$ denotes cumulative differences, $\boldsymbol{\varepsilon}_t = \left[\varepsilon_t^b \times (1 - F(z_{t-1})), \varepsilon_t^b \times F(z_{t-1})\right]'$ and $\boldsymbol{\beta}_h = \left[\beta_h^L, \beta_h^H\right]'$ are $2 \times 1$ vectors. Similarly, we gather all (lagged) control variables in $\boldsymbol{Z}_{t-1} = \left[(1, \boldsymbol{X}_{t-1}) \times (1 - F(z_{t-1})), (1, \boldsymbol{X}_{t-1}) \times F(z_{t-1}), t, t^2\right]'$ and define the corresponding coefficient vector $\boldsymbol{\gamma} = \left[\alpha_h^L, \boldsymbol{\gamma}_h^L, \alpha_h^H, \boldsymbol{\gamma}_h^H, \tau_{1h}, \tau_{2h}\right]'$, both as $k \times 1$ vectors where $k = 2(n_x + 1) + 2$.

Lusompa (2023) noted that the error term $u_{t+h}$ follows an autocorrelation process, which is known. Given that the data $\{y_t\}$ are stationary and purely non-deterministic,

such that there exists a Wold representation $y_t = \eta_t + \sum_{j=1}^{\infty} \Theta_j \eta_{t-j}$, Lusompa (2023) shows that the autocorrelation process in the error terms is known as

$$u_{t+h}^{(h)} = \Theta_h \eta_t + \ldots + \Theta_1 \eta_{t+h-1} + \eta_{t+h}, \tag{A.9}$$

which implies that there exists a linear and time-invariant vector moving average representation (VMA) of the uncorrelated one-step-ahead forecast errors $\{\eta_t\}$. In the population, the error process is a VMA($h$) even if the true model is not a (V)AR. Furthermore, it holds that

$$\phi_{1h} = \Theta_h \quad \Longrightarrow \quad u_{t+h}^{(h)} = \phi_{1h}\eta_t + \ldots + \phi_{11}\eta_{t+h-1} + \eta_{t+h}, \tag{A.10}$$

where $\phi_{1h} \in \boldsymbol{\gamma}_h$ corresponds to the coefficient of the first lag of the endogenous variable, $\Delta y_{t-1}$. Lusompa (2023) proposes using the following transformation

$$\Delta^h \tilde{y}_{t+h} = \Delta^h y_{t+h} - \phi_{1h}\hat{\eta}_t - \ldots - \phi_{11}\hat{\eta}_{t+h-1}, \tag{A.11}$$

which eliminates the autocorrelation in the residuals. For an estimate of the residuals, $\hat{\eta}_t$, we note that for the LP with horizon $h = 0$, $\eta_t = u_{t+h}^{(h)}$ holds.

This transformation can also be used in conjunction with Bayesian estimation. In the Bayesian treatment, one needs to set up the likelihood and elicit prior densities. Due to LPs being standard linear regressions, we elicit well-known independent priors for linear regressions. For both coefficient vectors, $\boldsymbol{\beta}_h$ and $\boldsymbol{\gamma}_h$, we impose independent Gaussian priors. Note that we are interested in the treatment effect estimation of $\boldsymbol{\beta}$, where the number of control variables is potentially large relative to the number of observations. Hence, we use a regularization prior for $\boldsymbol{\gamma}$. For each element of $\boldsymbol{\beta}$, we do not impose any form of regularization and instead impose an uninformative Gaussian prior given by

$$\beta_h^x \sim \mathcal{N}\left(\underline{\mu}_{h,\beta}^x, \underline{V}_{h,\beta}^x\right), \quad \forall x \in \{L, H\}, \tag{A.12}$$

where $\underline{\mu}_{h,\beta}^x = 0$ and $\underline{V}_{h,\beta}^x = 100$. Hahn et al. (2018) show that the presence of a regularization prior can still introduce a bias in the treatment effects. This bias arises due to confounding and depends on the predictability of $\boldsymbol{Z}_{t-1}$ by $\boldsymbol{e}_t$. Given the exogeneity of the sentiment shock and the predeterminedness of $\boldsymbol{Z}_{t-1}$, we argue that this is not a major issue.

For $\boldsymbol{\gamma}$, we define the prior variances as $\underline{\boldsymbol{V}}_{h,\gamma} = \text{diag}(\underline{v_1}, \ldots, \underline{v_k})$ with $\underline{v_i} = \lambda^2\psi_i^2$ for $i = 1, \ldots, k$. We propose a hierarchical global-local shrinkage prior setup based on the horseshoe (HS) prior following Carvalho et al. (2010), which reads as follows

$$\boldsymbol{\gamma}_h \sim \mathcal{N}\left(\underline{\boldsymbol{\mu}_{h,\gamma}}, \underline{\boldsymbol{V}_{h,\gamma}}\right), \quad \lambda \sim C^+(0,1), \quad \psi_i \sim C^+(0,1) \tag{A.13}$$

where $C^+$ denotes the half-Cauchy distribution. The parameter $\lambda_r$ denotes the global shrinkage parameter of regime $r$, which exerts shrinkage on all coefficients. The parameter $\psi_{rk}$ allows for individual, coefficient-specific shrinkage. Both parameters are estimated, and we do not have to specify any hyperparameters. We use the procedure outlined in Makalic and Schmidt (2015) to sample from the corresponding conditional posterior densities. We center the prior distribution of $\boldsymbol{\gamma}$ on zero, although other options are feasible as well (e.g., unity on the coefficient corresponding to the first own lag to resemble the Minnesota prior in Bayesian LPs; see also Ferreira et al., 2023). For the deterministic terms, we do not impose any shrinkage and assume a zero mean and a large variance, i.e., $10^5$. Regarding the prior distribution of the variance term, $\sigma_h^2$, we impose a conjugate inverse-Gamma prior distribution

$$\sigma_h^2 \sim iG(\underline{c}, \underline{d}), \tag{A.14}$$

where we set $\underline{c} = 3$ and $\underline{d} = 1$ to be uninformative.

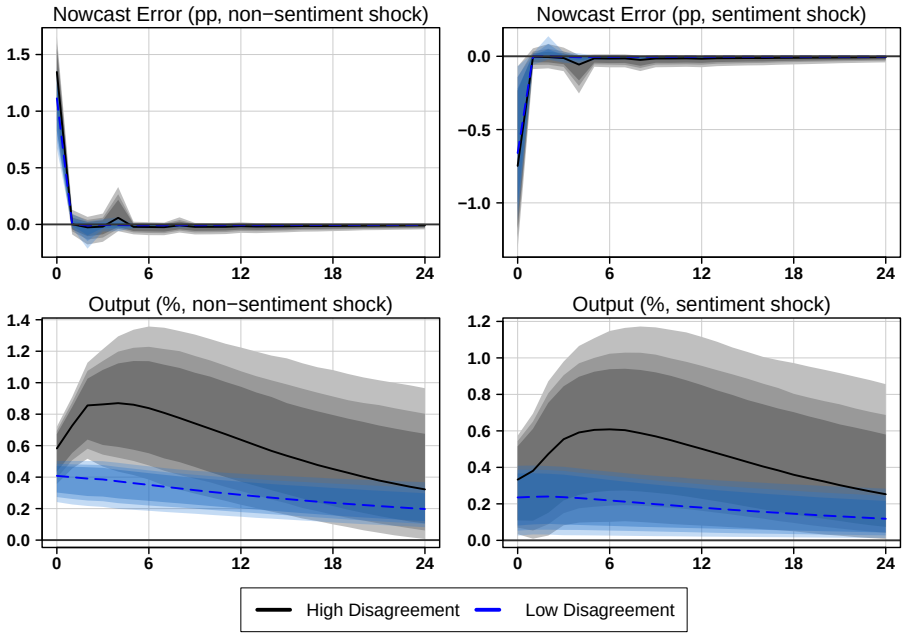
Similarly to the sampler of the STVAR, we briefly discuss the posterior simulation within an MCMC algorithm. Conditional on the regime indicator function $F(z_{t-1})$, the model is fully linear, and all conditional posterior distributions are available in closed form, rendering a Gibbs sampler convenient. Although the conditional posterior distributions of the VAR coefficients and the covariance matrix are standard, we use the auxiliary sampler of Makalic and Schmidt (2015) to sample from the conditional posterior densities for the HS prior. Lastly, if necessary, the parameter related to the transition function, $\theta$, has to be sampled with the help of a Metropolis-Hastings-within-Gibbs step since the posterior distribution is not available in closed form.

## B. Additional empirical results

This section reports additional empirical results not included in the main text. We split this section into three parts. First, we report any additional results regarding the outcomes of the smooth-transition vector autoregression. In the second part, we report additional results of the state-dependent local projections. In the third part, we provide additional checks for informational sufficiency.

### B.1 Additional results of the smooth-transition vector autoregression

**Figure B1:** Impulse responses by regime.



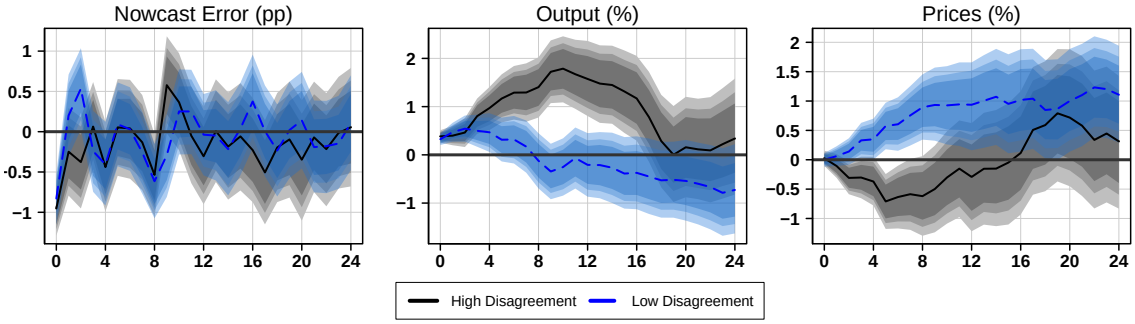

*Notes*: Estimates based on STVAR model. Identification based on sign-restrictions. The estimation covers the period 1969Q4-2019Q4. Black solid (blue dashed) line refers to the median response, while gray (blue) shaded areas indicate the 68%/80%/90% credible sets. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast errors) and percent (output).

Figure B1 reports the state-dependent impulse response functions of the STVAR model. We use the bivariate STVAR model to distinguish between sentiment and non-sentiment shocks. As is clearly visible in the figure, the non-sentiment shock shows positive comovement between the nowcast error and output, as imposed on impact through the sign restrictions. Similarly, we impose a negative comovement on impact between the nowcast error and output to identify a sentiment shock. In

terms of differences across regimes, we find stronger effects on output in the high-disagreement regime. However, the impulse response function is much more hump-shaped, obscuring the effects we observe in the local projections framework. This is due to the relatively small number of lags in the STVAR model, which may introduce a bias in the impulse responses beyond the horizon of the number of lags introduced in the STVAR.

### B.2 Additional results of the state-dependent local projections

**Figure B2:** Using the logistic function as the transition function.

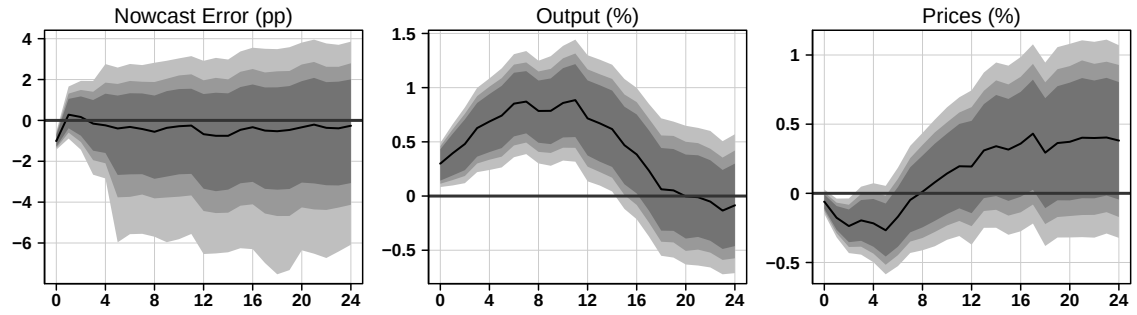



*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$) using a logistic transition function. Dependent variables are the nowcast error, output (log), and prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

In this section, we report a couple of additional results of the state-dependent local projections. First, we provide robustness to the type of transition function. While we use the empirical cumulative distribution function for our main results, we provide robustness by using the logistic function as the transition function. Therefore, we specify the transition function based on the logistic function, as discussed in Equation (A.4). We follow several previous studies that have calibrated rather than estimated the smoothness parameter $\theta$. We follow their suggestion and use a value of $\theta = 5$. Figure B2 reports the results, which show only negligible differences from the main outcomes in Figure 5.

In the next step, we account for the uncertainty of the generated regressor. As we pursue a Bayesian approach to estimation, we retrieve a full posterior distribution of the sentiment shock, which we denote as $p(\varepsilon_t^s)$. We adapt the algorithm for the Bayesian (ST)LP slightly. In each iteration of the Gibbs sampler, we draw from the

**Figure B3:** Accounting for the uncertainty of the generated regressor (LP).



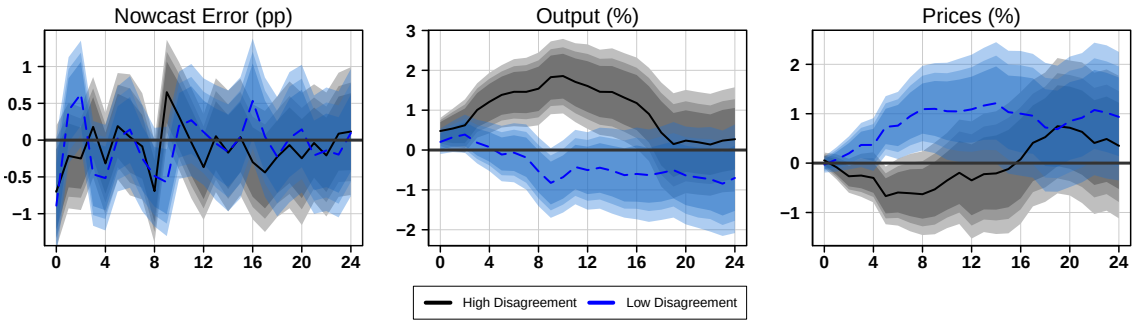

*Notes*: Estimates based on LP with an identified sentiment shock ($\varepsilon_t^s$) accounting also for the uncertainty around the sentiment shock. Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid lines refer to the median estimate; the gray areas refer to the 68%/80%/90% credible sets. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

distribution of the sentiment shocks: $(\varepsilon_t^s)^{(m)} \sim p(\varepsilon_t^s)$, where $(m)$ denotes the $m$-th iteration of the Gibbs sampler. This fully accounts for the uncertainty of the sentiment shock as a generated regressor when estimating the (ST)LP. We report the outcomes of this adjusted procedure in Figure B3 and Figure B4. While the posterior median is the same, credible sets have increased as we account for the additional uncertainty transmitted through the estimation of the sentiment shock itself. However, credible sets still indicate significant results, as in the baseline results. Hence, our results are robust when accounting for this uncertainty explicitly. This comes as no surprise since Wooldridge (2002, p. 117) notes that generated instruments do not suffer from the inference problem associated with generated regressors highlighted by Pagan (1984). We note that our sentiment shock series is not an instrument but an exogenous shock, but the conditions imposed are similar.

We report some additional robustness checks concerning the specification in Figure B5. We detrend the disagreement series by using the Hodrick-Prescott filter (instead of relying on scaling with the volatility of GDP growth), which is another approach to removing the low-frequency movements in the series. We have estimated the STLP also with a lower ($p = 2$) and higher ($p = 6$) number of lags. Lastly, we add more control variables to the STLP: we include the nowcast error, output, and prices in the set of controls. Our results are robust to these additional modifications.

We perform some additional robustness checks regarding the threshold variable. In the baseline, we use the interquartile range of the forecasters' forecasts in the SPF

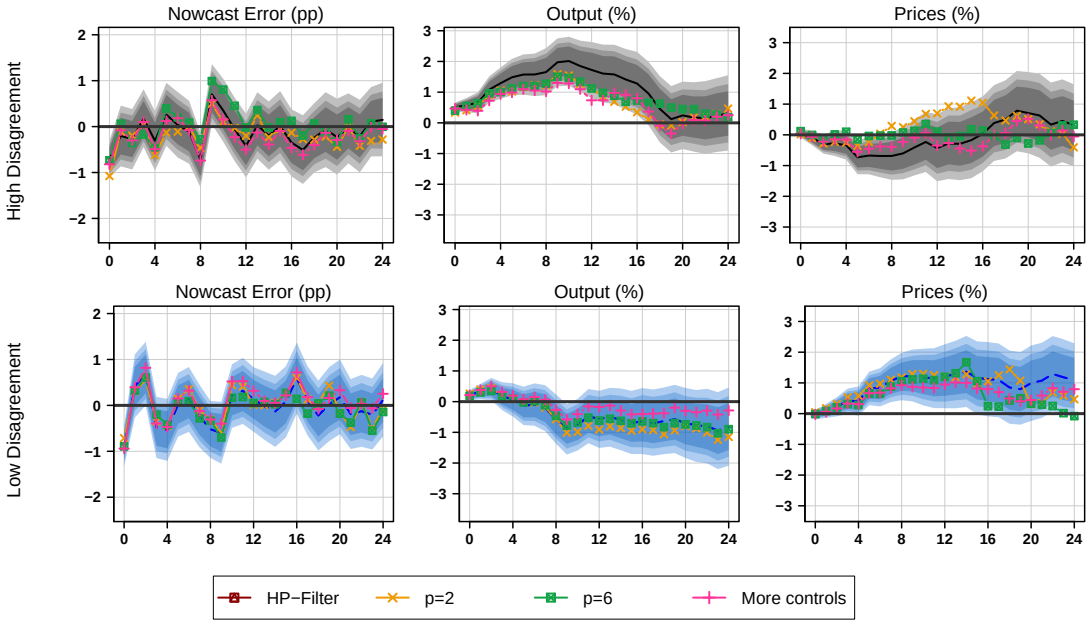
**Figure B4:** Accounting for the uncertainty of the generated regressor (STLP).




*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$) accounting also for the uncertainty around the sentiment shock. Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

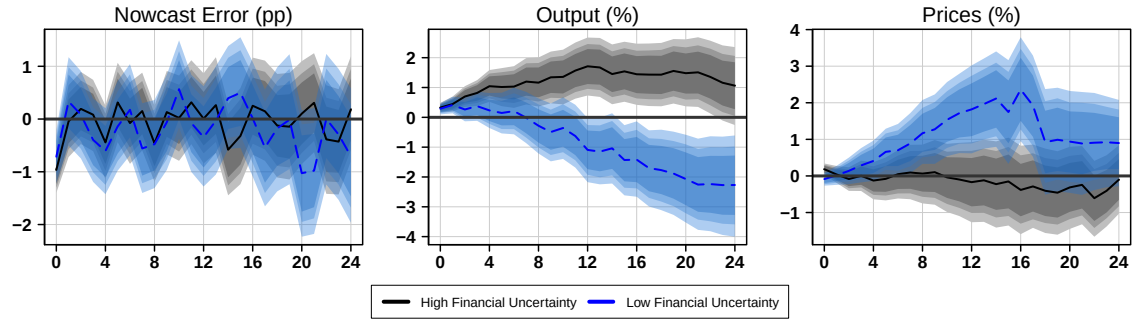
regarding output growth as a measure of disagreement. Additionally, we scale the series using the moving standard deviation of output growth with a window of 24 quarters. However, we could choose other threshold variables as well. Here, uncertainty measures come to mind, as effects may differ for disagreement and uncertainty measures (Born et al., 2023; Gambetti et al., 2023). We thus provide a robustness check in which we use the financial and macro uncertainty measures provided by Jurado et al. (2015). These measures are statistically significantly correlated with our baseline measure. The correlation coefficients are 0.35 ($p < 0.01$) and 0.22 ($p < 0.01$) for financial and macro uncertainty, respectively. As a final check, we scale our baseline disagreement series by the real-time moving standard deviation of output growth.

**Figure B5:** Additional robustness checks.

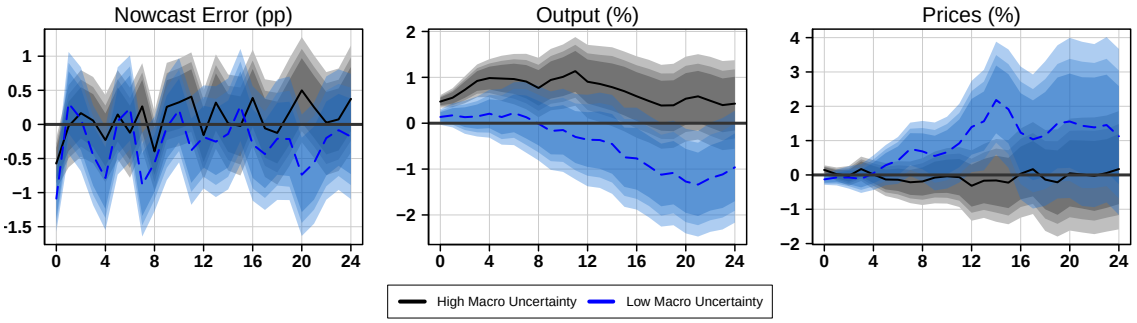



*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. Colored lines with symbols refer to robustness specifications: detrending disagreement with the Hodrick-Prescott filter, using fewer ($p = 2$) and more ($p = 6$) lags, and using more control variables. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

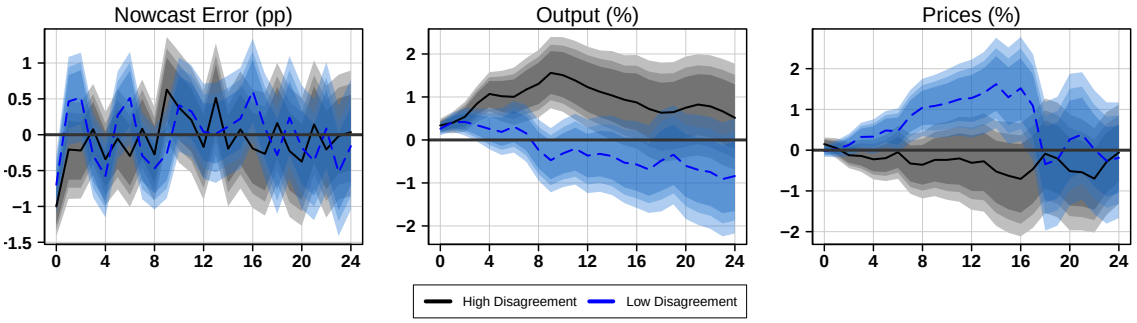
**Figure B6:** Additional robustness checks: uncertainty and real-time threshold variable.




**(a)** Threshold variable: Financial uncertainty.




**(b)** Threshold variable: Macro uncertainty.




**(c)** Threshold variable: Real-time disagreement.

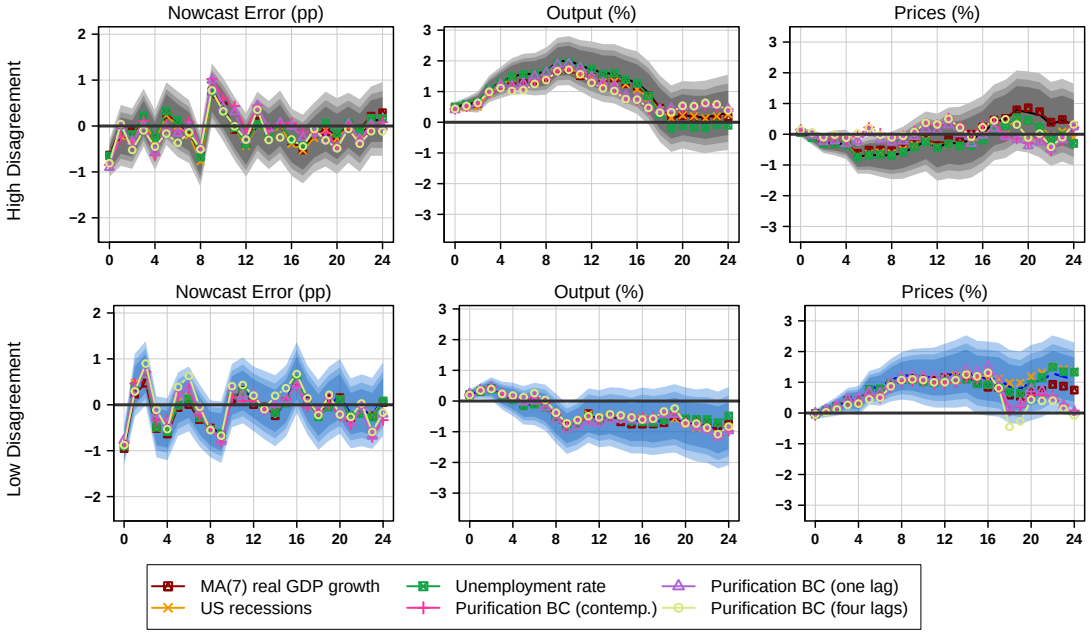
*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

We report the results of these checks in Figure B6. Our main findings are robust to these alternative indicators of uncertainty and disagreement. We find a stronger output reaction in the high-disagreement/high-uncertainty regime, while we find a stronger price reaction in the low-disagreement/low-uncertainty regime. In the high disagreement/uncertainty regime, prices are not statistically different from zero. For output in the low-disagreement regime, the outcomes are more mixed. When using disagreement, we find, similar to our main results, only a short-lived positive reaction that quickly becomes insignificant and fluctuates around zero. For the uncertainty measures, we even find a reversal of the initial positive reaction after three years. Although we think the comparison to uncertainty measures is important, we opt for disagreement in our baseline. The reason is mainly consistency: We construct both disagreement and the nowcast error using responses from the same survey. This allows us to investigate the effect of sentiment shocks conditional on the level of disagreement under which these forecasters operate. In contrast, the uncertainty measures are constructed outside of the SPF and thus may not ideally mimic the informational environment of these forecasters.

We conduct further robustness checks with respect to the state of the business cycle. As disagreement is correlated with the business cycle (Dovern et al., 2012), an alternative interpretation is that our results are not due to information frictions, but that capacity utilization differs over the business cycle. During recessions, a demand impulse boosts output, while during booms, the economy is near full capacity, and a demand impulse is absorbed by prices. First, we report low correlations with the state of the business cycle ($\rho = -0.06$ of the baseline disagreement measure with a seven-period moving average process of quarterly growth rates; $\rho = -0.14$ for year-on-year growth rates). This is in contrast to Dovern et al. (2012), who focus not on nowcasts for the disagreement but on one-year-ahead expectations for constructing disagreement.

Then, we conduct two exercises. In the first analysis, we re-estimate the baseline version of the model while additionally controlling for the level of the business cycle. We use three different measures: MA(7) of real GDP growth, a dummy for recessions, and the unemployment rate. This eliminates any potential level effects of the business cycle. In the second exercise, we off-project the information contained in the business cycle from the disagreement series before constructing the state indicator (*purification*). In this purification exercise, we use the MA(7) of real GDP growth for the business cycle. This orthogonalizes the disagreement series with respect to the business cycle.

**Figure B7:** Controlling for the state of the business cycle.

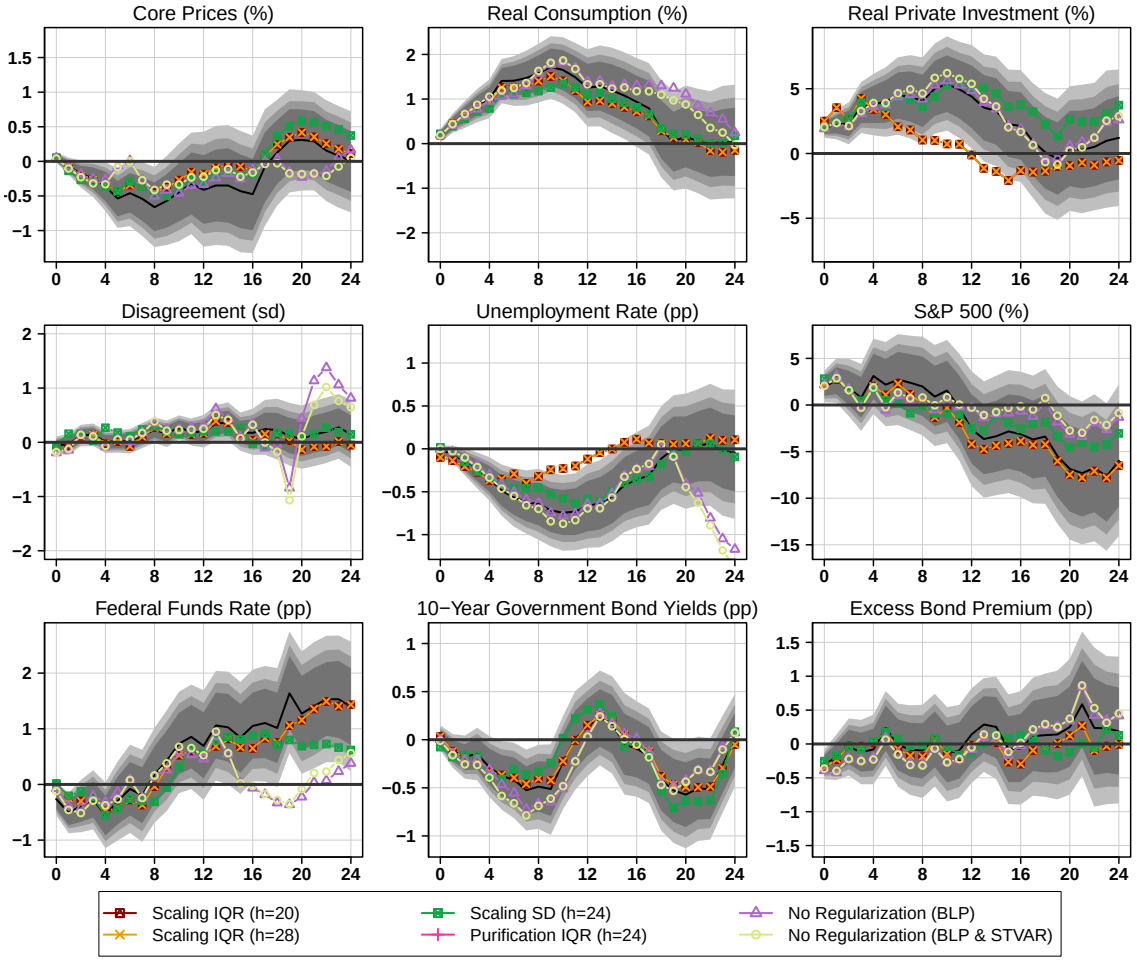



*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. Colored lines with symbols refer to robustness specifications: adding business cycle controls (MA(7) of real GDP growth, dummy for recessions, and the unemployment rate) and purification of the disagreement series by the state of the business cycle (only contemporaneous correlation or adding one or four lags). The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

Figure B7 reports the results, showing the median estimate and credible sets of the baseline model, as well as the median estimates of the alternative specifications. The results are robust both to controlling for the level of the business cycle and to purging the disagreement indicator of business-cycle information. None of the alternative models diverge strongly from the baseline model.

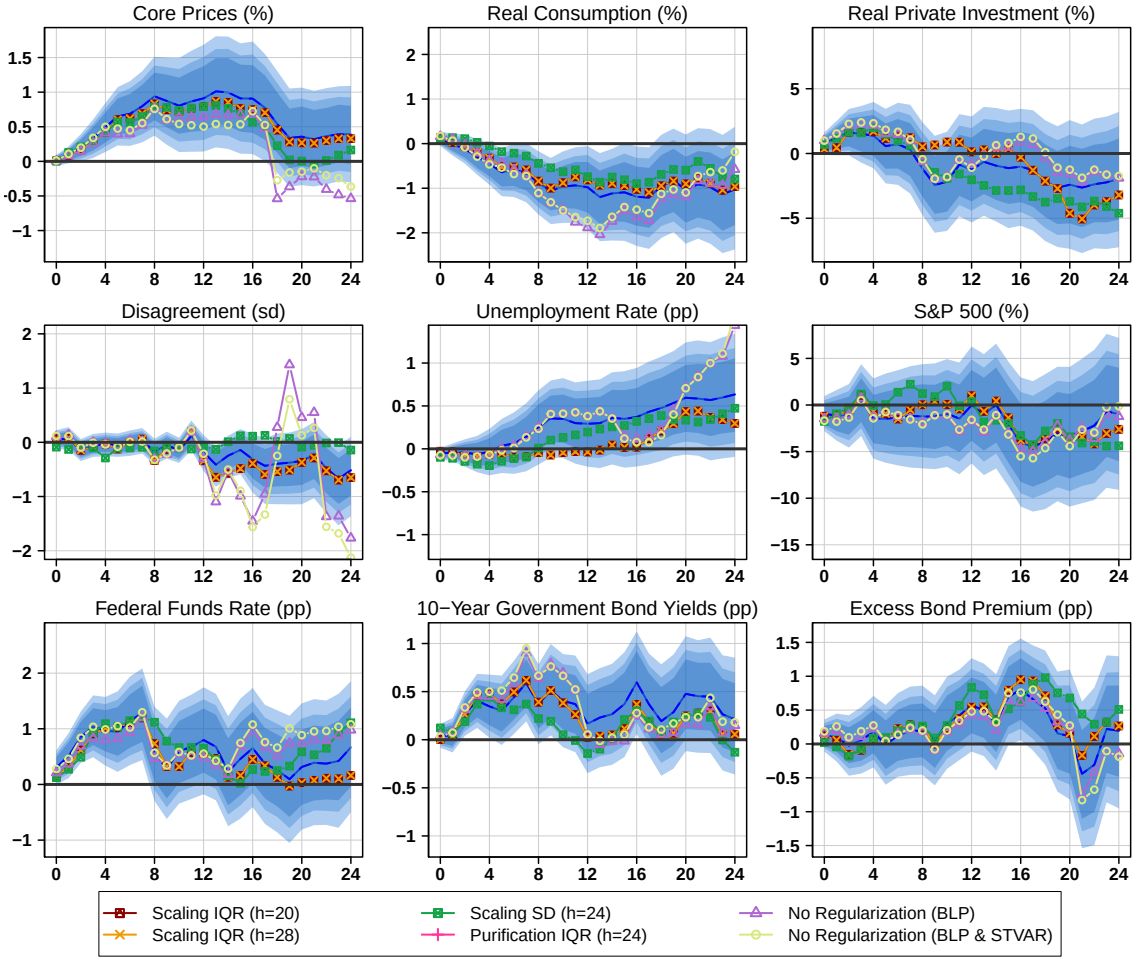
In Figure 9, we report robustness regarding the construction of the scaling variable for the state variable. We use the standard deviation instead of the interquartile range, or we perform purification. Furthermore, we change the window size when computing the moving standard deviation of GDP growth. We refer to purification as off-projecting the information contained in the moving standard deviation via linear regression. Additionally, we report robustness checks for using different prior distributions that impose no regularization for our main variables of interest (nowcast error, output, and prices), in contrast to our extended set of variables. Here, we report those additional variables in Figures B8 and B9.

**Figure B8:** Robustness to the state-dependent effects of a sentiment shock (high disagreement).



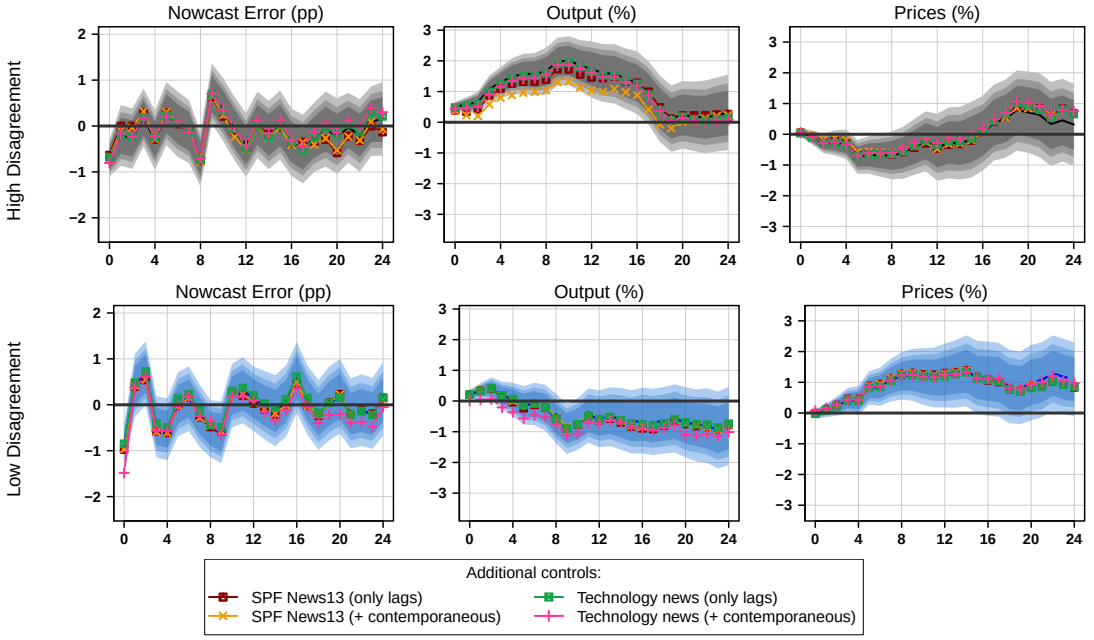

*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the core consumer prices (log), real consumption (log), real private investment (log), forecaster disagreement, the unemployment rate, the S&P 500 (log), the federal funds rate, the 10-year government bond yields, and the excess bond premium. The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. Colored lines with symbols refer to robustness specifications: Scaling using a different moving average process (20 or 28 quarters), scaling using the standard deviation as the dispersion measure, using purification instead of scaling, and prior sensitivity analysis (no regularization in the second or both estimation steps). The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percent (core consumer prices, real consumption, real investment, S&P 500), percentage points (unemployment rate, federal funds rate, government bond yield, excess bond premium), or standard deviations (disagreement).

**Figure B9:** Robustness to the state-dependent effects of a sentiment shock (low disagreement).




*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the core consumer prices (log), real consumption (log), real private investment (log), forecaster disagreement, the unemployment rate, the S&P 500 (log), the federal funds rate, the 10-year government bond yields, and the excess bond premium. The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. Colored lines with symbols refer to robustness specifications: Scaling using a different moving average process (20 or 28 quarters), scaling using the standard deviation as the dispersion measure, using purification instead of scaling, and prior sensitivity analysis (no regularization in the second or both estimation steps). The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percent (core consumer prices, real consumption, real investment, S&P 500), percentage points (unemployment rate, federal funds rate, government bond yield, excess bond premium), or standard deviations (disagreement).

**Figure B10:** Controlling for news and technology news shocks.




*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The estimation covers the period 1969Q4-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. Colored lines with symbols refer to robustness specifications: adding SPF news and the technology news shock as lagged controls or lagged and contemporaneous controls. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

We conduct an additional robustness check to investigate whether news about future output growth drives our results. Therefore, we also control for two different variables that capture news. We add these control variables either lagged or along with the contemporaneous value. Following the fiscal policy news literature (Caggiano et al., 2015), we construct the "News13" variable by summing revisions of expectations as follows:

$$\eta_{13}^y = \sum_{j=1}^{3} \left( \mathbb{E}_t \Delta y_{t+j} - \mathbb{E}_{t-1} \Delta y_{t+j} \right) \tag{B.1}$$

where $\mathbb{E}_t \Delta y_{t+j}$ is the median forecast of the growth rate of real output from period $t+j-1$ to period $t+j$ based on the information available at time $t$. Hence, $\mathbb{E}_t \Delta y_{t+j} - \mathbb{E}_{t-1} \Delta y_{t+j}$ represents the 'news' that becomes available to private agents between time $t-1$ and $t$ about the growth rate of real output $j$ periods ahead. We use the series DRGDP2 from the SPF, which we also use for the nowcasts. This measure

of anticipated output growth changes is intended to capture any additional news affecting output growth at longer horizons. Hence, it safeguards against concerns that expectations about future developments drive our results.

Furthermore, we use an identified technology news shock as an additional control variable. We use the identified news shock that maximizes the forecast error variance of total factor productivity at a long but finite horizon, which we have used in Section 4.

Figure B10 reports the results. It compares the baseline responses with specifications that separately include SPF forecast revisions or the identified technology news shock as additional controls, using either lagged values alone or both lagged and contemporaneous values. The posterior median responses remain close to the baseline, preserving the main differences across disagreement regimes. These findings suggest that the estimated response patterns are robust to conditioning on these measures of forecast revisions and technology news.

### B.3 Additional checks for informational sufficiency

We also assess the informational sufficiency of the baseline STVAR. Recovering structural shocks from the model's innovations requires an adequate information set, in addition to appropriate identifying restrictions. If the included variables fail to capture relevant predictive information, the identified shocks may remain predictable from past information outside the model. Such predictability provides evidence of *informational insufficiency* and motivates checking whether our findings are robust to expanding the model's information set.

Our baseline information set only contains two variables: the nowcast error and output. As this rather small information set may raise concerns about information sufficiency, we conduct additional robustness checks. We proceed in three steps: i) we test for informational insufficiency, ii) if necessary, we extend the information set of the VAR by augmenting it with factors from a high-dimensional macroeconomic dataset, and iii) we perform the testing procedure again on the structural shocks conditional on a larger information set. Hereby, we follow the suggestions in Forni and Gambetti (2014).

To assess whether the identified sentiment shock is predictable from a broader information set, we implement a shock-orthogonality diagnostic following Forni and Gambetti (2014). We regress the estimated shock on a constant and lagged principal-component factors extracted from the FRED-QD database (McCracken and Ng, 2020) and use an $F$-test of the joint null hypothesis that all coefficients on the lagged fac-

**Table B1:** Predictability of identified sentiment shocks

| | | Baseline | | FA: $q = 2$ | | FA: $q = 4$ | | FA: $q = 6$ | | FA: $q = 8$ | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Factors | Lags | $p$-value | $R^2$ | $p$-value | $R^2$ | $p$-value | $R^2$ | $p$-value | $R^2$ | $p$-value | $R^2$ |
| 2 | 2 | **0.022** | 0.059 | 0.880 | 0.006 | 0.924 | 0.005 | 0.854 | 0.007 | 0.884 | 0.006 |
| 2 | 4 | 0.081 | 0.073 | 0.978 | 0.011 | 0.934 | 0.016 | 0.920 | 0.017 | 0.895 | 0.019 |
| 2 | 8 | **0.040** | 0.140 | 0.342 | 0.093 | 0.420 | 0.087 | 0.408 | 0.088 | 0.274 | 0.099 |
| 4 | 2 | 0.068 | 0.076 | 0.565 | 0.036 | 0.942 | 0.015 | 0.861 | 0.021 | 0.939 | 0.016 |
| 4 | 4 | 0.265 | 0.100 | 0.897 | 0.051 | 0.987 | 0.033 | 0.976 | 0.037 | 0.989 | 0.032 |
| 4 | 8 | 0.228 | 0.196 | 0.415 | 0.174 | 0.801 | 0.136 | 0.783 | 0.138 | 0.823 | 0.133 |
| 8 | 2 | **0.009** | 0.164 | 0.165 | 0.111 | 0.301 | 0.097 | 0.248 | 0.102 | 0.618 | 0.073 |
| 8 | 4 | **0.010** | 0.266 | 0.186 | 0.202 | 0.385 | 0.177 | 0.427 | 0.173 | 0.928 | 0.115 |
| 8 | 8 | **0.020** | 0.440 | 0.110 | 0.397 | 0.383 | 0.350 | 0.392 | 0.349 | 0.760 | 0.302 |

*Notes:* Each regression includes a constant and the indicated number of lagged FRED-QD factors. The table reports $p$-values from conventional $F$-tests of the joint null hypothesis that all lagged-factor coefficients are zero assuming homoskedastic, serially uncorrelated regression errors, together with regression $R^2$. Baseline denotes the shock identified from the two-variable model; FA denotes factor augmentation, with $q$ indicating the number of factors appended to the STVAR. The factors used as predictors are the same across model specifications. All regressions use 191 observations. Bold $p$-values indicate rejection at the 5% significance level, based on unrounded values. No adjustment for multiple testing is applied. Inference treats estimated shocks and factors as given. Non-rejection does not establish invertibility.

tors are zero. Principal-component factors provide a parsimonious summary of the common variation in large macroeconomic datasets, allowing a broad information set to be incorporated with relatively few additional variables, following Stock and Watson (2002). We consider alternative factor counts and lag lengths, using a common estimation sample across specifications, and report the corresponding $p$-values and regression $R^2$. Rejection indicates that past factor information predicts the identified shock, providing evidence against informational sufficiency relative to the selected information set; non-rejection does not establish invertibility. Inference uses conventional homoskedastic $F$-tests and treats the estimated shocks and factors as given, without accounting for their estimation uncertainty. Note that we work with the posterior mean of the identified structural sentiment shock.

We report the results in Table B1 in the two columns referring to "Baseline". The null hypothesis of orthogonality to lagged factors is rejected at the 5% level in five of the nine specifications, including all specifications using eight factors. These results indicate that the baseline sentiment shock is predictable from information contained in the broader macroeconomic panel, providing evidence against informational sufficiency for this shock.

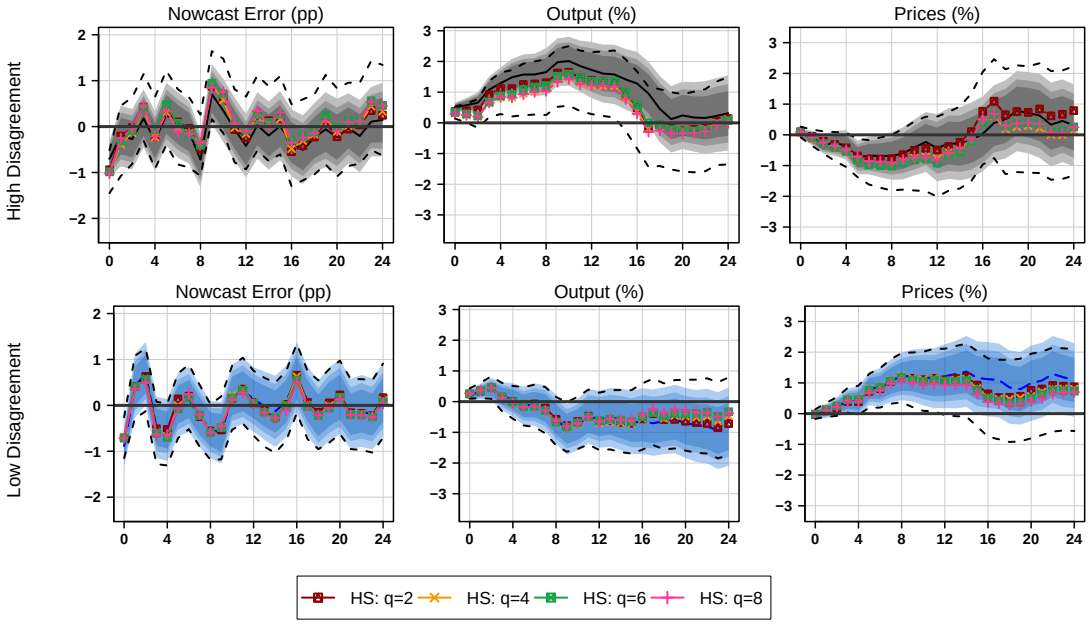
Motivated by these results, we augment the STVAR with the first $q$ principal-component factors from FRED-QD and re-estimate the model for $q \in \{2, 4, 6, 8\}$. The inclusion of factors slightly shortens the available estimation sample, so the STLPs using shocks from the factor-augmented STVARs are estimated over 1970Q2-2019Q4. Identification retains the baseline distinction between sentiment and non-sentiment shocks. In each regime, the sentiment shock is required to generate negative co-movement between output and the nowcast error on impact, while its effects on all factor variables are unrestricted. Each of the remaining $q + 1$ structural shocks is required to generate non-negative co-movement between output and the nowcast error on impact, again leaving all factor responses unrestricted. For the factor-augmented STVAR, the impact sign restrictions in each regime $r$ are summarized schematically as

$$\begin{bmatrix} u_t^{ne} \\ u_t^{gdp} \\ u_t^{F_1} \\ \vdots \\ u_t^{F_q} \end{bmatrix} = \begin{pmatrix} + & + & + & \cdots & + \\ + & - & + & \cdots & + \\ * & * & * & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ * & * & * & \cdots & * \end{pmatrix} \begin{bmatrix} \varepsilon_t^{ns} \\ \varepsilon_t^{s} \\ \varepsilon_t^{3} \\ \vdots \\ \varepsilon_t^{q+2} \end{bmatrix}, \tag{B.2}$$

where the entries of the impact matrix indicate sign restrictions rather than coefficient values, and $*$ denotes an unrestricted entry. No sign restrictions are imposed at subsequent horizons. The remaining shocks are not assigned separate economic interpretations, as they share the same sign restrictions.

We then re-estimate the STLP using the sentiment shocks identified from the factor-augmented STVARs. Figure B11 shows that the state-dependent responses remain robust to expanding the information set. The posterior median responses remain close to the baseline estimates, although the effects are somewhat weaker: under high disagreement, the output response peaks after ten quarters at approximately 1.4–1.6%, compared with 2.0% in the baseline. The dashed lines summarize uncertainty across the factor-augmented specifications by tracing, at each horizon, the minimum lower and maximum upper posterior quantiles. Moreover, the columns corresponding to the factor-augmented models in Table B1 show that the null of orthogonality to lagged factors is not rejected at the 5% level in any specification. Thus, the tests no longer detect evidence of informational insufficiency for the sentiment shock relative to the factor information considered, although non-rejection does not establish invertibility.

**Figure B11:** Enlarging the information set.




*Notes*: Estimates based on STLP with an identified sentiment shock ($\varepsilon_t^s$). Dependent variables are the nowcast error, output measured by real GDP (log), and prices measured by consumer prices (log). The factor-augmented specifications are estimated over the period 1970Q2-2019Q4. The black solid (blue dashed) lines refer to the median estimates of $\beta_h^H$ ($\beta_h^L$) in the high-disagreement (low-disagreement) regime. The gray (blue) areas refer to the 68%/80%/90% credible sets of the respective regime. Colored lines with symbols show posterior median responses using sentiment shocks identified from STVARs augmented with $q \in \{2, 4, 6, 8\}$ factors. Black dashed bounds trace the minimum lower (using the 5th) and maximum upper posterior (using the 95th) quantiles across these specifications. The horizontal axis measures the impulse response horizon in quarters. The vertical axis denotes deviations from trend in percentage points (nowcast error) or percent (output, prices).

## C. Extensions to the signal-extraction problem

In this section, we extend the signal-extraction problem introduced in Section 2 by considering alternative sources of changes in the information environment. Rather than varying the volatility of the fundamental, we study changes in the precision and relative composition of private and public signals.

As in Section 2, agents do not directly observe the fundamental $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. Instead, agent $i$ receives a private signal $a_i = \varepsilon + \eta_i$ with $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$, where the idiosyncratic noise $\eta_i$ averages to zero across agents. Agents also observe the public signal $s = \varepsilon + e$ with $e \sim \mathcal{N}(0, \sigma_e^2)$. All shocks are mutually independent.

The optimal estimate of the fundamental is

$$\mathbb{E}_i[\varepsilon \mid s, a_i] = \rho s + \delta a_i = (\rho + \delta)\varepsilon + \rho e + \delta \eta_i, \tag{C.1}$$

where

$$\rho = \frac{\sigma_\eta^2}{\sigma_e^2 + \sigma_\eta^2 + \sigma_e^2\sigma_\eta^2/\sigma_\varepsilon^2}, \qquad \delta = \frac{\sigma_e^2}{\sigma_e^2 + \sigma_\eta^2 + \sigma_e^2\sigma_\eta^2/\sigma_\varepsilon^2}. \tag{C.2}$$

We distinguish two objects. First, *disagreement* is the cross-sectional variance of agents' posterior means,

$$D \equiv \mathbb{V}_i\left(\mathbb{E}_i[\varepsilon \mid s, a_i]\right) = \delta^2\sigma_\eta^2 = \frac{\sigma_\varepsilon^4\sigma_e^4\sigma_\eta^2}{\left(\sigma_\varepsilon^2\sigma_\eta^2 + \sigma_\varepsilon^2\sigma_e^2 + \sigma_\eta^2\sigma_e^2\right)^2}. \tag{C.3}$$

Since $\varepsilon$ and $e$ are common across agents, cross-sectional variation in posterior means is entirely due to $\eta_i$. Second, *uncertainty* refers to the variance of an individual agent's posterior belief about the fundamental, or equivalently, the variance of the agent's forecast error:

$$U \equiv \mathbb{V}_i(\varepsilon \mid s, a_i) = \frac{\sigma_\varepsilon^2\sigma_e^2\sigma_\eta^2}{\sigma_\varepsilon^2\sigma_\eta^2 + \sigma_\varepsilon^2\sigma_e^2 + \sigma_e^2\sigma_\eta^2}. \tag{C.4}$$

The distinction between $D$ and $U$ is important. A deterioration in the information available to agents unambiguously makes their posterior beliefs less precise but need not necessarily make these beliefs more dispersed across agents. We also examine sensitivity to the public signal, measured by its weight $\rho$ in the posterior mean. Holding the fundamental and the private signal fixed, $\partial\mathbb{E}_i[\varepsilon \mid s, a_i]/\partial s = \rho$. Equivalently, $\rho$ measures the response of expectations to a unit realization of the public noise shock $e$, holding the other shocks fixed. The ensuing propositions characterize how this sensitivity changes alongside uncertainty and disagreement.

**Proposition 3 (Private-signal noise)** *An increase in private-signal noise unambiguously increases posterior uncertainty and sensitivity to the public signal,*

$$\frac{\partial U}{\partial \sigma_\eta^2} > 0, \qquad \frac{\partial \rho}{\partial \sigma_\eta^2} > 0,$$

*but increases disagreement if and only if*

$$\sigma_\eta^2 < \frac{\sigma_e^2 \sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_e^2}. \tag{C.5}$$

*Hence, disagreement is non-monotonic in private-signal noise.*

When private signals are initially sufficiently precise, making them noisier raises disagreement. Although the quality of private information deteriorates, agents continue to place substantial weight on their private signals, such that the additional idiosyncratic noise translates into greater dispersion of posterior beliefs. Once private information is sufficiently noisy, however, agents increasingly disregard it and rely instead on the common prior and public information. Further deterioration in private information then reduces disagreement, even though posterior uncertainty continues to increase. In the limit, as $\sigma_\eta^2 \to \infty$, $\delta \to 0$, and $D \to 0$.

The results of Section 5 also hold for an increase in private-signal noise if Condition (C.5) is satisfied. That is, the price response to noise shocks falls while the output response rises. However, given the strong evidence in Comin and Philippon (2005), Franco and Philippon (2007), Lorenzoni (2009), Andrade et al. (2022), or Dovern et al. (2023) that the volatility of idiosyncratic technology shocks is larger than that of aggregate ones, this condition is likely to be not fulfilled.

**Proposition 4 (Public-signal noise)** *An increase in public-signal noise unambiguously increases both posterior uncertainty and disagreement, but unambiguously reduces sensitivity to the public signal:*

$$\frac{\partial U}{\partial \sigma_e^2} > 0, \qquad \frac{\partial D}{\partial \sigma_e^2} > 0, \qquad \frac{\partial \rho}{\partial \sigma_e^2} < 0.$$

The intuition differs from the case of private-signal noise. As public information becomes less precise, agents optimally place relatively more weight on their heterogeneous private signals. Since these signals differ across agents, posterior beliefs become more dispersed. A deterioration in public information, therefore, raises both individual uncertainty and cross-sectional disagreement. In particular, this result shows that the positive mapping from a deterioration in the information environment to disagreement does not require an increase in the volatility of the fundamental.

Since the sensitivity to the public signal falls, we can rule out this possibility as an explanation for our results.

**Proposition 5 (Composition of signal noise)** *Define total signal-noise variance as* $\Omega = \sigma_e^2 + \sigma_\eta^2$*, and let* $\alpha \in (0,1)$ *denote the share of total signal noise attributable to the public signal, such that* $\sigma_e^2 = \alpha\Omega$ *and* $\sigma_\eta^2 = (1-\alpha)\Omega$*. Holding total signal noise* $\Omega$ *fixed, an increase in* $\alpha$ *shifts noise from private toward public information. This increases disagreement whenever*

$$0 < \alpha \leq \frac{2}{3}.$$

*In particular, if the public signal is at least as precise as the private signal,* $\sigma_e^2 \leq \sigma_\eta^2$*, shifting signal noise from private toward public information increases disagreement. For all* $0 < \alpha < 1$*, this shift reduces sensitivity to the public signal:*

$$\frac{\partial \rho}{\partial \alpha} < 0.$$

The result illustrates that both the overall quality and the composition of information affect disagreement. Holding the total amount of signal noise fixed, shifting noise toward the common signal makes agents rely relatively more strongly on heterogeneous private information and therefore tends to increase disagreement. The stated condition is naturally satisfied if the producer of the public signal—for instance, a statistical agency or media outlet—collects at least as much information as an individual agent, such that the public signal is initially more precise than the average private signal. However, since the sensitivity to the public signal falls, we can rule out this possibility as an explanation for our results.

**Summary.** Taken together, the results show that posterior uncertainty, disagreement, and sensitivity to the public signal are distinct objects whose relationship depends on the underlying change in the information environment. Greater private-signal noise increases posterior uncertainty and public-signal sensitivity but raises disagreement only when private information is sufficiently precise. This is unlikely to be satisfied. Greater public-signal noise, by contrast, increases uncertainty and disagreement while reducing public-signal sensitivity. Likewise, shifting noise from private toward public information reduces public-signal sensitivity, even when it increases disagreement. These results qualify the mechanism discussed in Section 2: Higher fundamental uncertainty jointly increases disagreement and sensitivity to sentiment shocks. The discussed alternatives fail to achieve these effects or require conditions that are very unlikely to hold to generate this result.

### C.1 Proofs

**Proof of Proposition 3.** Differentiating Equation (C.4) with respect to $\sigma_\eta^2$ gives

$$\frac{\partial U}{\partial \sigma_\eta^2} = \frac{\sigma_\varepsilon^4 \sigma_e^4}{\left(\sigma_\varepsilon^2\sigma_e^2 + \sigma_\varepsilon^2\sigma_\eta^2 + \sigma_e^2\sigma_\eta^2\right)^2} > 0.$$

Thus, making private information less precise always raises posterior uncertainty. Differentiating the public-signal weight in Equation (C.2) gives

$$\frac{\partial \rho}{\partial \sigma_\eta^2} = \frac{\sigma_\varepsilon^4 \sigma_e^2}{\left(\sigma_\varepsilon^2\sigma_\eta^2 + \sigma_\varepsilon^2\sigma_e^2 + \sigma_e^2\sigma_\eta^2\right)^2} > 0.$$

Differentiating equation (C.3) instead gives

$$\frac{\partial D}{\partial \sigma_\eta^2} = \frac{\sigma_e^4 \sigma_\varepsilon^4 \left[\sigma_e^2\sigma_\varepsilon^2 - \sigma_\varepsilon^2\sigma_\eta^2 - \sigma_e^2\sigma_\eta^2\right]}{\left(\sigma_e^2\sigma_\varepsilon^2 + \sigma_\eta^2\sigma_\varepsilon^2 + \sigma_e^2\sigma_\eta^2\right)^3}.$$

The denominator is strictly positive. Therefore, $\partial D/\partial \sigma_\eta^2 > 0$ if and only if

$$\sigma_e^2\sigma_\varepsilon^2 > \sigma_\eta^2\left(\sigma_\varepsilon^2 + \sigma_e^2\right),$$

which yields condition (C.5). ■

**Proof of Proposition 4.** Differentiating the public-signal weight in Equation (C.2) gives

$$\frac{\partial \rho}{\partial \sigma_e^2} = -\frac{\sigma_\varepsilon^2 \sigma_\eta^2(\sigma_\varepsilon^2 + \sigma_\eta^2)}{\left(\sigma_\varepsilon^2\sigma_\eta^2 + \sigma_\varepsilon^2\sigma_e^2 + \sigma_e^2\sigma_\eta^2\right)^2} < 0.$$

Differentiating Equation (C.4) with respect to $\sigma_e^2$ gives

$$\frac{\partial U}{\partial \sigma_e^2} = \frac{\sigma_\varepsilon^4 \sigma_\eta^4}{\left(\sigma_\varepsilon^2\sigma_e^2 + \sigma_\varepsilon^2\sigma_\eta^2 + \sigma_e^2\sigma_\eta^2\right)^2} > 0.$$

Similarly, differentiating disagreement gives

$$\frac{\partial D}{\partial \sigma_e^2} = \frac{2\sigma_e^2 \sigma_\varepsilon^6 \sigma_\eta^4}{\left(\sigma_e^2\sigma_\varepsilon^2 + \sigma_\eta^2\sigma_\varepsilon^2 + \sigma_e^2\sigma_\eta^2\right)^3} > 0.$$

■

**Proof of Proposition 5.** Under the stated parameterization, the public-signal weight is

$$\rho(\alpha) = \frac{\sigma_\varepsilon^2(1-\alpha)}{\sigma_\varepsilon^2 + \alpha(1-\alpha)\Omega}.$$

Hence,

$$\frac{\partial \rho}{\partial \alpha} = -\frac{\sigma_\varepsilon^2 \left[\sigma_\varepsilon^2 + (1-\alpha)^2\Omega\right]}{\left[\sigma_\varepsilon^2 + \alpha(1-\alpha)\Omega\right]^2} < 0.$$

Substituting $\sigma_e^2 = \alpha\Omega$ and $\sigma_\eta^2 = (1-\alpha)\Omega$ into equation (C.3) gives

$$D(\alpha) = \frac{\sigma_\varepsilon^4 \alpha^2 (1-\alpha)\Omega}{\left[\sigma_\varepsilon^2 + \alpha(1-\alpha)\Omega\right]^2}.$$

Differentiating with respect to $\alpha$ yields

$$\frac{\partial D}{\partial \alpha} = \frac{\sigma_\varepsilon^4 \Omega \alpha \left[(2-3\alpha)\sigma_\varepsilon^2 + \alpha^2(1-\alpha)\Omega\right]}{\left[\sigma_\varepsilon^2 + \alpha(1-\alpha)\Omega\right]^3}. \tag{C.6}$$

For $0 < \alpha \leq 2/3$, all terms are strictly positive, such that

$$\frac{\partial D}{\partial \alpha} > 0.$$

If $\sigma_e^2 \leq \sigma_\eta^2$, then $\alpha \leq 1/2$, so the sufficient condition is satisfied. ■

## D. Model solution

In Appendix E, we provide the proofs for Propositions 1-2 in Section 5. In a preliminary step, we outline the model solution and key equilibrium relationships in Appendix D. Throughout, we consider a linear approximation to the equilibrium conditions of the model. Lower-case letters indicate percentage deviations from steady state.

We solve the model by backward induction. That is, we start by deriving inflation expectations regarding period $t+1$. Using the result in the Euler equation of the third stage of period $t$ allows us to determine price-setting decisions during stage two. Eventually, we obtain the short-run responses of aggregate variables to unexpected productivity shocks or to sentiment shocks.

**Expectations regarding period $t$+1.** Below, $\mathbb{E}_{k,t}$ stands for either $\mathbb{E}_{j,l,t}$, referring to the information set of producer $j$ on island $l$ at the time of her pricing decision, or for $\mathbb{E}_{l,t}$, referring to the information set of the household on island $l$ at the time of its consumption decision. Variables with only time subscripts refer to economy-wide values. The wage in period $t+1$ is set according to the expected aggregate labor supply

$$\mathbb{E}_{k,t}\varphi l_{t+1} = \mathbb{E}_{k,t}(w_{t+1} - p_{t+1} - c_{t+1}).$$

This equation is combined with the aggregated production function

$$\mathbb{E}_{k,t} y_{t+1} = \mathbb{E}_{k,t}(x_{t+1} + \alpha l_{t+1}),$$

the expected aggregate labor demand

$$\mathbb{E}_{k,t}(w_{t+1} - p_{t+1}) = \mathbb{E}_{k,t}[x_{t+1} + (1-\alpha) l_{t+1}],$$

and market clearing $y_{t+1} = c_{t+1}$ to obtain $\mathbb{E}_{k,t} x_{t+1} = \mathbb{E}_{k,t} y_{t+1} = \mathbb{E}_{k,t} c_{t+1}$. Furthermore, the expected Euler equation, together with the Taylor rule, is

$$\mathbb{E}_{k,t} c_{t+1} = \mathbb{E}_{k,t}(c_{t+2} + \pi_{t+2} - \psi \pi_{t+1}).$$

Agents expect the economy to be in a new steady state tomorrow ($\mathbb{E}_{k,t} c_{t+1} = \mathbb{E}_{k,t} c_{t+2}$), given the absence of state variables other than technology, which follows a unit root process. Ruling out explosive paths yields

$$\mathbb{E}_{k,t} \pi_{t+2} = \mathbb{E}_{k,t} \pi_{t+1} = 0.$$

**Stage three of period *t*.** After prices are set, each household observes $n$ prices in the economy. Since the productivity signal is public, the productivity level $a_{j,l,t} = a_{l,t}$—which is the same for all producers $j \in [0,1]$ on island $l$—can be inferred from each price $p_{j,l,t}$ of the good from producer $j$ on island $l$. Hence, household $l$ forms its expectations about the change in aggregate productivity according to

$$\mathbb{E}_{l,t} \Delta x_t = \rho_x^h s_t + \delta_x^h \hat{a}_{l,t},$$

where $\hat{a}_{l,t}$ is the average over the realizations of $a_{m,t} - x_{t-1}$ for each location $m$ in household $l$'s sample. The coefficients $\rho_x^h$ and $\delta_x^h$ are equal across households and depend on $n, \sigma_e^2, \sigma_\varepsilon^2$, and $\sigma_\eta^2$ in the following way:

$$\rho_x^h = \frac{\sigma_\eta^2/n}{\underbrace{\sigma_e^2 + \sigma_\eta^2/n + \frac{\sigma_e^2 \sigma_\eta^2/n}{\sigma_\varepsilon^2}}_{\to 0 \text{ if } n \to \infty}}, \qquad \delta_x^h = \frac{\sigma_e^2}{\underbrace{\sigma_e^2 + \sigma_\eta^2/n + \frac{\sigma_e^2 \sigma_\eta^2/n}{\sigma_\varepsilon^2}}_{\to 1 \text{ if } n \to \infty}}. \tag{D.1}$$

Producers, on the other hand, only observe the signal and their own productivity. They thus form expectations according to

$$\mathbb{E}_{j,l,t} \Delta x_t = \rho_x^p s_t + \delta_x^p (a_{l,t} - x_{t-1}), \tag{D.2}$$

with

$$\rho_x^p = \frac{\sigma_\eta^2}{\sigma_e^2 + \sigma_\eta^2 + \frac{\sigma_\eta^2 \sigma_e^2}{\sigma_\varepsilon^2}} \qquad \delta_x^p = \frac{\sigma_e^2}{\sigma_e^2 + \sigma_\eta^2 + \frac{\sigma_\eta^2 \sigma_e^2}{\sigma_\varepsilon^2}},$$

such that $\delta_x^h > \delta_x^p$ because of the higher information content of households' observations. Consumption follows an Euler equation with household-specific inflation, as only a subset of goods is bought. Agents expect no differences between households for $t+1$, such that expected aggregate productivity and the overall price level impact today's individual consumption. Also using $\mathbb{E}_{l,t}p_{t+1} = \mathbb{E}_{l,t}p_t$ and $\mathbb{E}_{l,t}x_{t+1} = \mathbb{E}_{l,t}x_t$ gives

$$c_{l,t} = \mathbb{E}_{l,t}x_t + \mathbb{E}_{l,t}p_t - p_{l,t} - r_t. \tag{D.3}$$

Similar to the updating formula for technology estimates, households use their available information to form an estimate about the aggregate price level $p_t$ according to

$$\mathbb{E}_{l,t}p_t = \rho_p^h s_t + \delta_p^h \hat{a}_{l,t} + \kappa_p^h w_t + \tau_p^h x_{t-1} - \eta_p^h r_t. \tag{D.4}$$

Combining the above gives

$$c_{l,t} = (1+\tau_p^h)x_{t-1} + \rho_{xp}^h s_t + \delta_{xp}^h \hat{a}_{l,t} + \kappa_p^h w_t - (1+\eta_p^h)r_t - p_{l,t}, \tag{D.5}$$

where $\rho_{xp}^h = \rho_x^h + \rho_p^h$ and $\delta_{xp}^h = \delta_x^h + \delta_p^h$. We will solve for the undetermined coefficients below.

**Stage two of period $t$.** During the second stage, firms obtain idiosyncratic signals about their productivity. In the following, the index $\tilde{p}_{l,t}$ is the average price index of customers visiting island $l$. If customers bought on all (that is, infinitely many) islands in the economy, $\tilde{p}_{l,t}$ would correspond to the overall price level. Since consumers only buy on a subset of islands, the price of their own island has a non-zero weight in their price index, which is taken into account further below. Firms set prices according to

$$\begin{aligned} p_{j,l,t} &= w_t + \frac{1-\alpha}{\alpha}\mathbb{E}_{j,l,t}y_{j,l,t} - \frac{1}{\alpha}a_{l,t} \\ &\equiv k' + k_1'\mathbb{E}_{j,l,t}\tilde{p}_{l,t} + k_2'\mathbb{E}_{j,l,t}y_t - k_3'a_{l,t}, \end{aligned}$$

with

$$k' = \frac{\alpha}{\alpha+\gamma(1-\alpha)}w_t \quad k_1' = \frac{\gamma(1-\alpha)}{\alpha+\gamma(1-\alpha)} \quad k_2' = \frac{1-\alpha}{\alpha+\gamma(1-\alpha)} \quad k_3' = \frac{1}{\alpha+\gamma(1-\alpha)}. \tag{D.6}$$

From here onwards, expressions that are based on common knowledge only (such as $k'$) are treated like parameters in notation terms, i.e. they lack a time index. This facilitates the important distinction between expressions that are common information

and those that are not. Evaluating the expectation of firm $j$ about aggregate output in period $t$, given equation (D.5), results in

$$\mathbb{E}_{j,l,t}y_t = \kappa^h + \rho^h_{xp}s_t + \delta^h_{xp}\mathbb{E}_{j,l,t}\left(\frac{1}{n}a_{l,t} + \frac{n-1}{n}\mathbb{E}_{j,l,t}x_t - x_{t-1}\right) - \left(\frac{1}{n}p_{j,l,t} + \frac{n-1}{n}\mathbb{E}_{j,l,t}p_t\right),$$

where $\kappa^h = (1+\tau^h_p)x_{t-1} - (1+\eta^h_p)r_t + \kappa^h_p w_t$ contains only publicly available information. Furthermore, it is taken into account that the productivity of island $l$ has a non-zero weight in the sample of productivity levels observed by consumers visiting island $l$. Note that producers still take the price index of the consumers as given, since consumers buy infinitely many goods on the same island. Inserting the above into the pricing equation (D.6) yields (here, $p_t$ is the average of the prices charged by producers of all other islands, which is the overall price index as there are infinitely many locations)

$$p_{j,l,t} \equiv k + k_1\mathbb{E}_{j,l,t}p_t + \tilde{k}s_t - k_3 a_{l,t},$$

with

$$\Xi = 1 - \frac{1}{n}(k'_1 - k'_2) \qquad k = \frac{1}{\Xi}\left\{k' + k'_2\kappa^h + \frac{k'_2\delta^h_{xp}}{n}\left[(n-1)(1-\delta^p_x) - 1\right]x_{t-1}\right\} \quad \text{(D.7)}$$

$$k_1 = \frac{n-1}{n\Xi}\left(k'_1 - k'_2\right) \qquad \tilde{k} = \frac{k'_2}{\Xi}\left(\rho^h_{xp} + \delta^h_{xp}\rho^p_x\frac{n-1}{n}\right) \qquad k_3 = \frac{1}{\Xi}\left\{k'_3 - \frac{k'_2\delta^h_{xp}}{n}\left[(n-1)\delta^p_x + 1\right]\right\}.$$

Note that, according to (D.6), $0 < k'_1 - k'_2 < 1$ because $0 < \alpha < 1$ and $\gamma > 1$. Using the definition of $k_1$ in (D.7), this implies (observe that $n > 1$)

$$0 < k_1 < 1.$$

Aggregating over all producers gives the aggregate price index

$$p_t = k + k_1\overline{E}_t p_t + \tilde{k}s_t - k_3 x_t,$$

where $\int a_{l,t}dl = x_t$, and $\overline{E}_t p_t = \iint \mathbb{E}_{j,l,t}p_t \; djdl$ is the average expectation of the price level.

The expectation of firm $j$ of this aggregate is therefore

$$\begin{aligned}\mathbb{E}_{j,l,t}p_t &= k + \tilde{k}s_t - k_3\mathbb{E}_{j,l,t}x_t + k_1\mathbb{E}_{j,l,t}\overline{E}_t p_t \\ &= k + \left(\tilde{k} - k_3\rho^p_x\right)s_t - k_3\delta^p_x a_{l,t} - k_3(1-\delta^p_x)x_{t-1} + k_1\mathbb{E}_{j,l,t}\overline{E}_t p_t. \qquad \text{(D.8)}\end{aligned}$$

Inserting the last equation into (D.7) gives

$$p_{j,l,t} = k + k_1 k - k_1 k_3(1-\delta^p_x)x_{t-1} + \left[\tilde{k} + k_1\left(\tilde{k} - k_3\delta^p_x\right)\right]s_t - \left(k_3 + k_1 k_3\delta^p_x\right)a^j_t + k_1^2\mathbb{E}_{j,l,t}\overline{E}_t p_t.$$

To find $\mathbb{E}_{j,l,t}\overline{E}_t p_t$, note that firm $j$'s expectations of the average of (D.8) are

$$\mathbb{E}_{j,l,t}\overline{E}_t p_t = k - k_3(1-\delta_x^p)(1+\delta_x^p)x_{t-1} + \left(\tilde{k} - k_3\rho_x^p - k_3\delta_x^{p2}\rho_x^p\right)s_t - k_3\delta_x^{p2}a_{l,t} + k_1\mathbb{E}_{j,l,t}\overline{E}_t^{(2)}p_t,$$

where $\overline{E}^{(2)}$ is the average expectation of the average expectation. The price of firm $j$ is found by plugging the last equation into the second-to-last:

$$\begin{aligned} p_{j,l,t} = &\left(k + k_1 k + k_1^2 k\right) - \left[k_1 k_3(1-\delta_x^p) + k_1^2 k_3(1-\delta_x^p)(1+\delta_x^p)\right]x_{t-1} \\ &+ \left[\tilde{k} + k_1\left(\tilde{k} - k_3\rho_x^p\right) + k_1^2\left(\tilde{k} - k_3\rho_x^p - k_3\delta_x^p\rho_x^p\right)\right]s_t \\ &- \left(k_3 + k_1 k_3\delta_x^p + k_1^2 k_3\delta_x^{p2}\right)a_{l,t} + k_1^3\mathbb{E}_{j,l,t}\overline{E}^{(2)}p_t. \end{aligned}$$

Continuing like this results in some infinite sums

$$\begin{aligned} p_{j,l,t} =& k\left(1 + k_1 + k_1^2 + k_1^3\ldots\right) \\ &- k_1 k_3(1-\delta_x^p)\Big[1 + k_1(1+\delta_x^p) + k_1^2(1+\delta_x^p+\delta_x^{p2}) + k_1^3(1+\delta_x^p+\delta_x^{p2}+\delta_x^{p3}\ldots)\Big]x_{t-1} \\ &+ \left[\tilde{k} + k_1\left(\tilde{k} - k_3\rho_x^p\right) + k_1^2\left(\tilde{k} - k_3\rho_x^p - k_3\delta_x^p\rho_x^p\right) + k_1^3\left(\tilde{k} - k_3\rho_x^p - k_3\rho_x^p\delta_x^p - k_3\rho_x^p\delta_x^{p2}\right) + \ldots\right]s_t \\ &- k_3\left(1 + k_1\delta_x^p + k_1^2\delta_x^{p2} + k_1^3\delta_x^{p3}\ldots\right)a_{l,t} + k_1^\infty\mathbb{E}_{j,l,t}\overline{E}^{(\infty)}p_t. \end{aligned}$$

For the terms in the third line, we have

$$\begin{aligned} &\tilde{k} + k_1\left(\tilde{k} - k_3\rho_x^p\right) + k_1^2\left(\tilde{k} - k_3\rho_x^p - k_3\delta_x^p\rho_x^p\right) + k_1^3\left(\tilde{k} - k_3\rho_x^p - k_3\rho_x^p\delta_x^p - k_3\rho_x^p\delta_x^{p2}\right) \\ &+ k_1^4\left(\tilde{k} - k_3\rho_x^p - k_3\rho_x^p\delta_x^p - k_3\rho_x^p\delta_x^{p2} - k_3\rho_x^p\delta_x^{p3}\right)\ldots \\ =&\tilde{k}(1 + k_1 + k_1^2 + k_1^3\ldots) - \left(k_1 k_3\rho_x^p + k_1^2 k_3\rho_x^p + k_1^3 k_3\rho_x^p\ldots\right) \\ &- \left(\delta_x^p k_1^2 k_3\rho_x^p + \delta_x^p k_1^3 k_3\rho_x^p + \delta_x^p k_1^4 k_3\rho_x^p\ldots\right) - \left(\delta_x^{p2} k_1^3 k_3\rho_x^p + \delta_x^{p2} k_1^4 k_3\rho_x^p + \delta_x^{p3} k_1^5 k_3\rho_x^p\ldots\right)\ldots \\ =&\tilde{k}(1 + k_1 + k_1^2 + k_1^3\ldots) - k_1 k_3\left(\frac{\rho_x^p}{1-k_1} + \frac{\rho_x^p\delta_x^p k_1}{1-k_1} + \frac{\rho_x^p\delta_x^{p2}k_1^2}{1-k_1}\ldots\right) \\ =&\frac{\tilde{k}}{1-k_1} - \frac{k_1 k_3\rho_x^p}{1-k_1}\left(1 + \delta_x^p k_1 + \delta_x^{p2}k_1^2\ldots\right) \\ =&\frac{\tilde{k}}{1-k_1} - \frac{k_1 k_3\rho_x^p}{(1-k_1)(1-\delta_x^p k_1)}. \end{aligned}$$

Proceeding similarly with the terms in the other lines results in

$$p_{j,l,t} = \frac{k}{1-k_1} - \frac{k_1(1-\delta_x^p)}{1-k_1}\frac{k_3}{1-k_1\delta_x^p}x_{t-1} + \frac{1}{1-k_1}\left(\tilde{k} - \rho_x^p\frac{k_1 k_3}{1-k_1\delta_x^p}\right)s_t - \frac{k_3}{1-k_1\delta_x^p}a_{l,t} + \underbrace{k_1^\infty\overline{E}_t^{(\infty)}}_{\to 0}p_t.$$

Setting idiosyncratic technology shocks equal to zero in order to track the effects of aggregate shocks and observing that all firms then set the same price gives

$$p_t \equiv \bar{k}_1 + \bar{k}_2 s_t + \bar{k}_3 x_t,$$

with

$$\bar{k}_1 = \frac{1}{1-k_1}\left[k - (1-\delta_x^p)\frac{k_1 k_3}{1-k_1\delta_x^p}x_{t-1}\right] \quad \bar{k}_2 = \frac{1}{1-k_1}\left(\tilde{k} - \rho_x^p \frac{k_1 k_3}{1-k_1\delta_x^p}\right) \quad \bar{k}_3 = -\frac{k_3}{1-k_1\delta_x^p}. \tag{D.9}$$

To arrive at qualitative predictions for the impact of the structural shocks $\varepsilon_t$ and $e_t$ on output growth and the nowcast error, we need to determine the sign and the size of $\bar{k}_3$. Note that, according to (D.7),

$$-k_3 = \delta_{xp}^h \frac{k_2' - nk_3'/\delta_{xp}^h + k_2'(n-1)\delta_x^p}{n-(k_1'-k_2')},$$

where the first part of the numerator can be rewritten, by observing (D.6), as

$$k_2' - nk_3'/\delta_{xp}^h = \frac{1 - n/\delta_{xp}^h - \alpha}{\alpha + \gamma(1-\alpha)}.$$

Using (D.6) and (D.7) thus yields

$$-k_3 = \delta_{xp}^h \frac{(1-\alpha)[(n-1)\delta_x^p + 1] - n/\delta_{xp}^h}{(n-1)[\alpha+\gamma(1-\alpha)]+1}.$$

Plugging this into the definition of $\overline{k}_3$ in (D.9) gives

$$\overline{k}_3 = \delta_{xp}^h \frac{\frac{(1-\alpha)[(n-1)\delta_x^p+1]-n/\delta_{xp}^h}{(n-1)[\alpha+\gamma(1-\alpha)]+1}}{1-\delta_x^p\frac{(n-1)(\gamma-1)(1-\alpha)}{(n-1)[\alpha+\gamma(1-\alpha)]+1}}.$$

To obtain $\delta_{xp}^h = \delta_x^h + \delta_p^h$, we need to find the undetermined coefficients of equation (D.4). Start by comparing this equation with household $l$'s expectation of equation (D.9):

$$\mathbb{E}_{l,t}p_t = \underbrace{\overline{k}_1 + \overline{k}_3 x_{t-1}}_{\kappa_p^h w_t + \tau_p^h x_{t-1} - \eta_p^h r_t} + \underbrace{\left(\overline{k}_2 + \overline{k}_3\rho_x^h\right)}_{\rho_p^h} s_t + \underbrace{\overline{k}_3\delta_x^h}_{\delta_p^h}\hat{a}_{l,t}. \tag{D.10}$$

Hence, $\delta_{xp}^h = \delta_x^h(1+\overline{k}_3)$. Inserting this into the above expression for $\overline{k}_3$ yields

$$\overline{k}_3 \equiv -\frac{n/\Upsilon - \delta_x^h\Psi}{\Phi - \delta_x^h\Psi}, \tag{D.11}$$

with

$$\Upsilon = (n-1)[\alpha+\gamma(1-\alpha)]+1 > 0 \qquad \Psi = (1-\alpha)[(n-1)\delta_x^p+1]/\Upsilon > 0$$
$$\Phi = 1 - \delta_x^p(n-1)(\gamma-1)(1-\alpha)/\Upsilon.$$

The signs obtain because $n > 1, 0 < \alpha < 1, \delta_x^p > 0$, and $\gamma > 1$. Observe that $\Psi\Upsilon < n$ because $\delta_x^p \leq 1$. Hence, $n/\Upsilon - \delta_x^h\Psi > 0$ because

$$n - \underbrace{\delta_x^h}_{>0,<1}\underbrace{\Psi\Upsilon}_{<n} > 0,$$

implying that the numerator of (D.11) is positive. Turning to the denominator $\Phi - \delta_x^h \Psi$, observe that $\Phi - \Psi > 0$. The denominator of (D.11) is therefore positive as well, and we have $\overline{k}_3 < 0$. Next, consider that $n/\Upsilon < \Phi$ and we obtain

$$-1 < \overline{k}_3 < 0.$$

This is a key result for the derivation of Propositions 1 and 2; see Appendix E. Multiplying the numerator and the denominator of the fraction in equation (D.11) by $\Upsilon$ and rewriting gives the expression used in Proposition 1.

**Prices** We now investigate the effect of a noise shock on prices, that is, we set the other shocks to zero. Using equation (D.9), we get

$$p_t = \bar{k}_1 + \bar{k}_2 e_t,$$

where $\bar{k}_1$ includes the wage. Furthermore,

$$\bar{k}_1 = \frac{k}{1-k_1} \qquad \bar{k}_2 = \frac{1}{1-k_1}\left[\tilde{k} + \underbrace{k_1 \overline{k}_3 \rho_x^p}_{>-1,<0}\right],$$

where we have used equation (D.9). According to (D.7),

$$k = \frac{1}{\Xi}\left[k' + k_2' \kappa_p^h w_t\right].$$

Remember that

$$k' = \frac{\alpha}{\alpha + \gamma(1-\alpha)} w_t \qquad k_2' = \frac{1-\alpha}{\alpha + \gamma(1-\alpha)} \qquad \Xi = 1 - \frac{1}{n}(k_1' - k_2'),$$

see (D.6) and (D.7). As a result,

$$\Xi = \frac{n\alpha + (1-\alpha)[(n-1)\gamma + 1]}{n\alpha + n\gamma(1-\alpha)}.$$

Note that $1 - k_1$ can be written as

$$1 - k_1 = \frac{n}{(n-1)[\alpha + \gamma(1-\alpha)] + 1}.$$

Taken together, this gives

$$k = \frac{1}{\Xi} \frac{1}{\alpha + \gamma(1-\alpha)} \left[\alpha + (1-\alpha)\kappa_p^h\right] w_t \tag{D.12}$$

$$\rightarrow \bar{k}_1 = \frac{k}{1-k_1} = \left[\alpha + (1-\alpha)\kappa_p^h\right] w_t. \tag{D.13}$$

According to (D.10)

$$\bar{k}_1 = \kappa_p^h w_t.$$

Combining the last two equations yields

$$\bar{k}_1 = w_t \quad \rightarrow \quad \kappa_p^h = 1.$$

Turning to $\bar{k}_2$, observe that according to (D.7)

$$\tilde{k} = \frac{1}{\Xi}\frac{1-\alpha}{\alpha+\gamma(1-\alpha)}\left[\rho_{xp}^h + \delta_{xp}^h\frac{n-1}{n}\rho_x^p\right].$$

Hence, keeping in mind the derivation of (D.13),

$$\bar{k}_2 = \frac{1}{1-k_1}\left[\tilde{k} + k_1\overline{k}_3\rho_x^p\right] = (1-\alpha)\left[\rho_{xp}^h + \delta_{xp}^h\frac{n-1}{n}\rho_x^p\right] + \frac{k_1}{1-k_1}\overline{k}_3\rho_x^p.$$

Inserting the insights from (D.10) yields

$$\begin{aligned}\bar{k}_2 &= (1-\alpha)\left[\rho_x^h + \rho_p^h + (\delta_x^h+\delta_p^h)\frac{n-1}{n}\rho_x^p\right] + \frac{k_1}{1-k_1}\overline{k}_3\rho_x^p \\ &= \frac{1-\alpha}{\alpha}\left[\rho_x^h(1+\bar{k}_3) + \delta_x^h(1+\overline{k}_3)\frac{n-1}{n}\rho_x^p\right] + \frac{k_1}{\alpha(1-k_1)}\overline{k}_3\rho_x^p.\end{aligned}$$

Then observe that, see (D.12),

$$\frac{1}{\alpha}\frac{k_1}{1-k_1} = \frac{n-1}{n}(\gamma-1)\frac{1-\alpha}{\alpha}.$$

The total effect of noise on inflation, according to

$$p_t = w_t + \bar{k}_2 e_t$$

is then (where $\Omega \equiv -\overline{k}_3$ is used in the main text)

$$\frac{\partial p_t}{\partial e_t} = (1+\overline{k}_3)\frac{1-\alpha}{\alpha}\left[\rho_x^h + \delta_x^h\frac{n-1}{n}\rho_x^p\right] + (\gamma-1)\frac{n-1}{n}\frac{1-\alpha}{\alpha}\overline{k}_3\rho_x^p, \tag{D.14}$$

**Stage one of period $t$** As information sets of agents are perfectly aligned during stage one, we use the expectation operator $\mathbb{E}_t$ to denote (common) stage-one expectations in what follows. Combining the results regarding expectations about inflation in period $t+1$ with the Euler equation, the Taylor rule, and the random-walk assumption for $x_t$ gives

$$\mathbb{E}_t y_t = \mathbb{E}_t x_t - \psi\mathbb{E}_t\pi_t.$$

Remember that the monetary policy shock emerges after wages are set. Its expected value before wage-setting is zero. Combining labor supply and demand with the production function then yields $\mathbb{E}_t\pi_t = 0$, such that $\mathbb{E}_t y_t = \mathbb{E}_t x_t = x_{t-1}$. Nominal wages are set in line with these expectations. We thus have determinacy of the price level. The central bank also expects zero inflation in the absence of monetary policy shocks.

## E. Proofs

**Proof of Proposition 1** Aggregating individual Euler equations (D.3) over all individuals, using (D.9) and (D.10),

$$\begin{aligned} y_t =& \mathbb{E}_{l,t}x_t + \mathbb{E}_{l,t}p_t - p_t - r_t \\ =& x_{t-1} + \rho_x^h(1+\overline{k}_3)s_t + \left[\delta_x^h + \overline{k}_3(\delta_x^h - 1)\right]\varepsilon_t - \frac{\alpha}{\alpha+\psi(1-\alpha)}\nu_t \\ =& x_{t-1} + \underbrace{\rho_x^h(1+\overline{k}_3)}_{>0} e_t + \underbrace{\left[\delta_x^h + \rho_x^h - \overline{k}_3(1-\delta_x^h-\rho_x^h)\right]}_{>0}\varepsilon_t - \underbrace{\frac{\alpha}{\alpha+\psi(1-\alpha)}}_{<0}\nu_t, \end{aligned} \tag{E.1}$$

where $1-\delta_x^h-\rho_x^h > 0$ because of (D.1). Note that, if households have full information ($n \to \infty$), we get $\rho_x^h \to 0$ and $\delta_x^h \to 1$. Defining $\Omega \equiv -\overline{k}_3$, we can write

$$y_t = x_{t-1} + \rho_x^h(1-\Omega)e_t + \left[(\delta_x^h + \rho_x^h)(1-\Omega) + \Omega\right]\varepsilon_t - \frac{\alpha}{\alpha+\psi(1-\alpha)}\nu_t.$$

The signs indicated above result from $0 < \Omega = -\overline{k}_3 < 1$ (derived in Appendix D).

Now consider the nowcast error, where expectations are either those of households or producers, that is, $\mathbb{E}_{k,t}$ substitutes for either $\mathbb{E}_{j,l,t}$ or $\mathbb{E}_{l,t}$, and $\rho^k, \delta^k$ correspondingly for $\rho^p, \delta^p$ or $\rho^h, \delta^h$. Taking expectations of equation (E.1) gives

$$\begin{aligned} \mathbb{E}_{k,t}y_t =& x_{t-1} + \rho_x^h\left(1+\overline{k}_3\right)s_t + \left[\delta_x^h + \overline{k}_3(\delta_x^h-1)\right]\mathbb{E}_{k,t}\varepsilon_t - r_t \\ =& x_{t-1} + \left\{\rho_x^h(1+\overline{k}_3) + [\delta_x^h + \overline{k}_3(\delta_x^h-1)]\rho_x^k\right\}s_t + \left[\delta_x^h + \overline{k}_3(\delta_x^h-1)\right]\delta_x^k\varepsilon_t - r_t. \end{aligned}$$

$$\begin{aligned} y_t - \mathbb{E}_{k,t}y_t =& -\rho_x^k\left[\delta_x^h + \overline{k}_3(\delta_x^h-1)\right]s_t + \left[\delta_x^h + \overline{k}_3(\delta_x^h-1)\right](1-\delta_x^k)\varepsilon_t \\ =& \underbrace{-\rho_x^k\left[\delta_x^h + \overline{k}_3(\delta_x^h-1)\right]}_{<0}e_t + \underbrace{\left[\delta_x^h + \overline{k}_3(\delta_x^h-1)\right]}_{>0}\underbrace{(1-\delta_x^k-\rho_x^k)}_{>0}\varepsilon_t, \end{aligned}$$

or

$$y_t - \mathbb{E}_{k,t}y_t = -\rho_x^k\left[\delta_x^h(1-\Omega)+\Omega\right]e_t + \left[\delta_x^h(1-\Omega)+\Omega\right](1-\delta_x^k-\rho_x^k)\varepsilon_t.$$

The fact that $0 < \Omega < 1$ allows us to determine the signs of the effects of the shocks on the nowcast error. ■

**Proof of Proposition 2** *A higher volatility $\sigma_\varepsilon^2$ of aggregate technology leads to...*

*...a higher dispersion of nowcasts and forecasts of output by firms*
As stated by equation (D.2), firms form their expectations according to

$$\mathbb{E}_{j,l,t}\Delta x_t = \rho_x^p s_t + \delta_x^p(a_{l,t} - x_{t-1}),$$

such that the dispersion of expectations is given by $(\delta_x^p)^2\sigma_\eta^2$. Remember that the public signal $s_t$ and the common productivity shock $\varepsilon_t$ are the same for all firms. The effect of $\sigma_\varepsilon^2$ on expectation dispersion is then

$$\frac{\partial(\delta_x^p)^2\sigma_\eta^2}{\partial\sigma_\varepsilon^2} = 2(\delta_x^p)^2\rho_x^p\frac{\sigma_\eta^2\sigma_e^2}{\sigma_\varepsilon^4},$$

which is positive, such that the impact of $\sigma_\varepsilon^2$ on the dispersion of nowcasts of $x_t$ is also positive. It follows that nowcasts and forecasts of inflation and output are more dispersed for a higher $\sigma_\varepsilon^2$.

*...a higher dispersion of nowcasts and forecasts of output by households*
The derivation for households is equivalent to that of firms, with the only difference that $\delta_x^h$ is used instead of $\delta_x^p$.

*...a higher impact of positive optimism shocks on output*
Equation (E.1) states

$$y_t = x_{t-1} + \rho_x^h(1+\overline{k}_3)e_t + \left[\delta_x^h + \rho_x^h - \overline{k}_3(1-\delta_x^h-\rho_x^h)\right]\varepsilon_t - \frac{\alpha}{\alpha+\psi(1-\alpha)}\nu_t.$$

Taking the derivative w.r.t. $e_t$ gives

$$\frac{\partial y_t}{\partial e_t} = \rho_x^h(1+\overline{k}_3). \tag{E.2}$$

We are then interested in whether the effect of dispersion, governed by $\sigma_\varepsilon^2$, on the above derivative is positive, that is whether

$$\frac{\partial\partial y_t/\partial e_t}{\partial\sigma_\varepsilon^2} > 0.$$

We proceed in two steps. First, we define, as in the main text, $\Omega = -\bar{k}_3$ for better readability and derive its derivative with respect to $\sigma_\varepsilon^2$. Writing $\Omega = N/D$ with

$$N = n - \delta_x^h(1-\alpha)\left[(n-1)\delta_x^p + 1\right] \quad D = n\alpha + (1-\alpha)\left\{(1-\delta_x^h)[1+\delta_x^p(n-1)] + (n-1)\gamma(1-\delta_x^p)\right\},$$

we have

$$\frac{\partial\Omega}{\partial\sigma_\varepsilon^2} = \frac{D\frac{\partial N}{\partial\sigma_\varepsilon^2} - N\frac{\partial D}{\partial\sigma_\varepsilon^2}}{D^2}.$$

Turning to the derivatives of the $\rho$ and $\delta$ coefficients, we define

$$\frac{\partial \delta_x^p}{\partial \sigma_\varepsilon^2} = \sigma_e^2/V^2 \qquad \frac{\partial \rho_x^p}{\partial \sigma_\varepsilon^2} = \sigma_\eta^2/V^2 \qquad \frac{\partial \delta_x^h}{\partial \sigma_\varepsilon^2} = n\sigma_e^2/W^2 \qquad \frac{\partial \rho_x^h}{\partial \sigma_\varepsilon^2} = \sigma_\eta^2/W^2,$$

with $V \equiv \sigma_\varepsilon^2 \left[\sigma_e^2 + \sigma_\eta^2 + \sigma_\eta^2\sigma_e^2/\sigma_\varepsilon^2\right]/\sigma_e\sigma_\eta$ and $W \equiv \sigma_\varepsilon^2 \left[\sigma_e^2 + \sigma_\eta^2/n + \sigma_\eta^2\sigma_e^2/(n\sigma_\varepsilon^2)\right] n/\sigma_e\sigma_\eta$. Using the definition $X \equiv [1 + (n-1)\delta_x^p] = W/V$ gives

$$\frac{\partial \Omega}{\partial \sigma_\varepsilon^2} = \frac{1-\alpha}{D^2}\sigma_e^2 \left\{(N-D)nX/W^2 - (n-1)\left[D\delta_x^h + N[(1-\delta_x^h) - \gamma]\right]/V^2\right\}.$$

Second, to derive the sign of

$$\frac{\partial \partial y_t/\partial e_t}{\partial \sigma_\varepsilon^2} = \frac{\partial \rho_x^h}{\partial \sigma_\varepsilon^2}(1-\Omega) - \rho_x^h \frac{\partial \Omega}{\partial \sigma_\varepsilon^2} = \sigma_\eta^2 \frac{D-N}{DW^2} - \rho_x^h \frac{\partial \Omega}{\partial \sigma_\varepsilon^2},$$

we define $\bar{\Pi} := (\partial Y/\partial \sigma_\varepsilon^2)V^2W^2D^2/\sigma_\eta^2$, such that

$$\bar{\Pi} = (D-N)\,V^2D - (1-\alpha)\,\rho_x^h\,\frac{V^2W^2\sigma_e^2}{\sigma_\eta^2}\left[(N-D)Xn/W^2 + (n-1)\left[N(\gamma - (1-\delta_x^h)) - D\delta_x^h\right]/V^2\right].$$

Now use the identities $\rho_x^h = \sigma_\eta\,\sigma_\varepsilon^2/(\sigma_e\,W)$ and

$$N[\gamma - (1-\delta_x^h)] - D\delta_x^h = (\gamma - 1)n\left[1 - (1-\alpha)\delta_x^h\right].$$

Substituting these and simplifying the common factors yields

$$\bar{\Pi} = (D-N)V^2D - (1-\alpha)\,\frac{\sigma_e}{\sigma_\eta}\,\sigma_\varepsilon^2\left[n(N-D)\,V + n(n-1)(\gamma-1)W\left(1-(1-\alpha)\delta_x^h\right)\right].$$

Note that

$$1 - \delta_x^p = \frac{\sigma_\eta(\sigma_e^2 + \sigma_\varepsilon^2)}{\sigma_e\,V} \qquad 1 - \delta_x^h = \frac{\sigma_\eta(\sigma_e^2 + \sigma_\varepsilon^2)}{\sigma_e\,W}$$

and, hence,

$$D = n\alpha + (1-\alpha)\,\frac{\sigma_\eta(\sigma_e^2 + \sigma_\varepsilon^2)}{\sigma_e\,V}\left[1 + (n-1)\gamma\right].$$

Therefore

$$\begin{aligned}\frac{\sigma_\eta(\sigma_e^2 + \sigma_\varepsilon^2)}{\sigma_e}VD &= \frac{\sigma_\eta(\sigma_e^2 + \sigma_\varepsilon^2)}{\sigma_e}V\left[n\alpha + (1-\alpha)\,\frac{\sigma_\eta(\sigma_e^2 + \sigma_\varepsilon^2)}{\sigma_e\,V}\left[1 + (n-1)\gamma\right]\right] \\ &= \frac{\sigma_e^2 + \sigma_\varepsilon^2}{\sigma_e^2}\Big\{[\sigma_\varepsilon^2\sigma_e^2 + \sigma_\varepsilon^2\sigma_\eta^2 + \sigma_\eta^2\sigma_e^2]n\alpha + \sigma_\eta^2(1-\alpha)(\sigma_e^2 + \sigma_\varepsilon^2)\left[1 + (n-1)\gamma\right]\Big\}.\end{aligned}$$

This can be used, together with $N - D = -(1-\alpha)(n-1)(1-\delta_x^p)(\gamma - 1)$, in the following derivations:

$$\begin{aligned}\bar{\Pi}/[(n-1)(\gamma-1)(1-\alpha)] &= (1-\delta_x^p)V^2D - \frac{\sigma_e}{\sigma_\eta}\,\sigma_\varepsilon^2\Big\{n[-(1-\alpha)(1-\delta_x^p)]V + nW\left(1 - \delta_x^h + \alpha\delta_x^h\right)\Big\} \\ &= \sigma_\eta^2\frac{(\sigma_e^2 + \sigma_\varepsilon^2)^2}{\sigma_e^2}\left\{n\alpha + (1-\alpha)\left[1 + (n-1)\gamma\right]\right\} - \alpha n^2\frac{\sigma_e^2\sigma_\varepsilon^4}{\sigma_\eta^2}.\end{aligned}$$

With this, one obtains, after collection of terms, a single scalar factor deciding the sign:

$$\bar{\Pi} = \frac{(1-\alpha)(n-1)(\gamma-1)}{\sigma_e^2\sigma_\eta^2}\alpha\left[Q\sigma_\eta^4(\sigma_e^2+\sigma_\varepsilon^2)^2 - n^2\sigma_e^4\sigma_\varepsilon^4\right],$$

where

$$Q = \{1+\gamma(n-1)(1-\alpha)+\alpha(n-1)\}/\alpha > 1.$$

Because the prefactor $[(1-\alpha)(n-1)(\gamma-1)]/(\sigma_e^2\sigma_\eta^2)$ is positive, we have

$$\text{sign}\Big(\frac{\partial\partial y_t/\partial e_t}{\partial\sigma_\varepsilon^2}\Big) = \text{sign}(\bar{\Pi}) = \text{sign}\Big(Q\sigma_\eta^4\,(\sigma_e^2+\sigma_\varepsilon^2)^2 - n^2\sigma_e^4\sigma_\varepsilon^4\,\Big).$$

That is, the effect of $\sigma_\varepsilon^2$ on the impact of noise shocks on GDP is positive if

$$Q\left(\frac{\sigma_\eta^2/n}{\sigma_\varepsilon^2}+\frac{\sigma_\eta^2/n}{\sigma_e^2}\right)^2 > 1.$$

Given that $Q > 1$, this inequality holds in realistic cases, particularly if

1. The volatility of the idiosyncratic technology shock is large, $\sigma_\eta^2 \to \infty$.
2. Households have a minimal informational advantage over firms, $n$ close to unity, and the volatility of the idiosyncratic technology shock is larger than the aggregate and/or noise volatility ($\sigma_\eta^2 > \sigma_\varepsilon^2$ or $\sigma_\eta^2 > \sigma_e^2$).

Alternatively, one can express this condition in terms of the signals' informational content. The condition does *not* hold if $\sigma_\varepsilon^2$ and $\sigma_e^2$ are both very large relative to $\sigma_\eta^2$. That is, the condition holds, except when the private signal of the households, composed of price observations, is very precise ($\sigma_\eta^2/n$ much smaller than $\sigma_\varepsilon^2$) and simultaneously contains much more information than the public signal ($\sigma_\eta^2/n$ much smaller than $\sigma_e^2$).

*...a lower impact of positive optimism shocks on prices*

Equation (D.14) gives the impact of noise on prices. Its derivative w.r.t. $\sigma_\varepsilon^2$ equals the effect of dispersion on the impact of noise on prices.

The derivative of $p_t$ with respect to $e_t$ is

$$\frac{\partial p_t}{\partial e_t} = (1-\Omega)\frac{1-\alpha}{\alpha}\left[\rho_x^h + \delta_x^h\frac{n-1}{n}\rho_x^p\right] - \Omega(\gamma-1)\frac{n-1}{n}\frac{1-\alpha}{\alpha}\rho_x^p.$$

If we define $K = \sigma_e^2+\sigma_\eta^2/n+\sigma_\eta^2\sigma_e^2/(n\sigma_\varepsilon^2)$ and $L = \sigma_e^2+\sigma_\eta^2+\sigma_\eta^2\sigma_e^2/(\sigma_\varepsilon^2)$, we can write

$$\frac{\partial p_t}{\partial e_t} = \frac{1-\alpha}{\alpha}\frac{\sigma_\eta^2}{nD}\Big\{\frac{D-N}{K}\left[L+(n-1)\sigma_e^2\right] - \frac{N}{L}(\gamma-1)(n-1)\Big\}.$$

Since $L + (n-1)\sigma_e^2 = nK$ and $D - N$ are given above, we have

$$\frac{\partial p_t}{\partial e_t} = \frac{1-\alpha}{\alpha}\frac{\sigma_\eta^2}{nDL}(\gamma-1)(n-1)\Big\{n(1-\alpha)(1-\delta_x^p) - N\Big\}.$$

Observing that $n - (n-1)\delta_x^h = L/K$, we can also write

$$\frac{\partial p_t}{\partial e_t} = -(\gamma-1)(n-1)(1-\alpha)\frac{\sigma_\eta^2}{DL} = -(\gamma-1)(n-1)(1-\alpha)\frac{\rho_x^p}{D}.$$

We know that $\rho_x^p$ increases in $\sigma_\varepsilon^2$, such that we only need the sign of the derivative of $D$. Note that

$$\begin{aligned} D &= n\alpha + (1-\alpha)\left\{\frac{K-\sigma_e^2}{K}\frac{nK}{L} + (n-1)\gamma\frac{L-\sigma_e^2}{L}\right\} \\ &= n\alpha + (1-\alpha)\frac{\sigma_\eta^2(1+\sigma_e^2/\sigma_\varepsilon^2)}{L}[1+(n-1)\gamma] \\ &= n\alpha + (1-\alpha)\left[1 + \frac{\sigma_e^2}{\sigma_\eta^2}\frac{1}{1+\sigma_e^2/\sigma_\varepsilon^2}\right]^{-1}[1+(n-1)\gamma], \end{aligned}$$

where we used $1 + \delta_x^p(n-1) = nK/L$. We hence obtain $\frac{\partial D}{\partial \sigma_\varepsilon^2} < 0$ and

$$\frac{\partial \partial p_t/\partial e_t}{\partial \sigma_\varepsilon^2} < 0.$$

■

## Appendix References


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